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DEPARTMENT OF DEFENSE HANDBOOK

ELECTROMAGNETIC ENVIRONMENTAL EFFECTS TO ORDNANCE GUIDE

PART 3

ELECTROSTATIC DISCHARGE TO ORDNANCE

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FOREWORD

1. This handbook is approved for use by all Departments or Agencies of the Department of Defense (DoD).

2. This part supplements MIL-HDBK-240-1 and was prepared for contractors, DoD test agencies, activities, and test planners. This handbook is for guidance only and cannot be cited as a requirement.

3. This handbook provides recommended practices for conducting Electromagnetic Environmental Effects (E3) to Ordnance evaluations. Developed over many years by DoD test activities, the methods and procedures described herein are the product of the collective DoD electrostatic discharge (ESD) test experience. This handbook consolidates these practices into a single document vice numerous individual Service standards, instructions, and so forth, thus promoting a standardized approach.

4. This handbook was prepared by the Services under the sponsorship of the Joint Ordnance Commander's Group (JOCG) E3 Ordnance Safety Subgroup in accordance with the guidelines of the Standardization Reform Policy established by the Secretary of Defense.

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1. SCOPE

1.1 Purpose. This part of MIL-HDBK-240 provides the procedures and techniques to conduct Personnel-borne Electrostatic Discharge (ESD) (PESD) and Helicopter-borne ESD (HESD) tests. It covers PESD and HESD testing for all Air Force, Army, Navy, Marine Corps, and Coast Guard ordnance and support equipment for all mission areas. Although intended primarily for use by Department of Defense (DoD) ESD test activities, this part of the handbook provides a consolidation of "corporate knowledge" about the subject that should be of interest to procurement authorities and system developers.

1.2 Background. There are three specific objectives of this handbook:

a. Harmonize the differences in the existing ESD test procedures used by each of the Services.

b. Reduce redundant testing.

c. Meet the ordnance safety requirements in MIL-STD-464.

The use of this standard should allow interoperability of ordnance facilitating the use or adaptation of Joint Service weapons. MIL-STD-464 provides the fundamental environment for PESD and HESD. Ordnance required to survive the MIL-STD-464 environment should be tested and evaluated using this standard. MIL-STD-331 serves as the fuze environmental performance standard for the qualification of fuzes, safe & arm (S&A) devices and ignition safety devices (ISD). MIL-STD-331 only addresses fuzes, where MIL-STD-464 and this handbook address the ordnance all-up round (AUR) and ordnance subcomponents. In order to facilitate the flow and acceptance of ESD reports, analyses, and data it is important that calibration, procedures, and sensitivity data conform to this document.

ESD test data are used to determine the susceptibility of ordnance and weapon systems to PESD and HESD. This information is used to assess risks and develop effective control measures to minimize those risks.

The PESD procedure applies to all categories of ordnance, such as ammunition, small arms, missiles, rockets, and pyrotechnics.

The HESD procedure applies to all categories of ordnance in the following configurations: bare item carry, the external carry (installed on aircraft), the hot tube loaded (install/uninstall on operating ungrounded aircraft), and the Vertical Replenishment (VERTREP)/sling load carry (packaged configuration).

2. APPLICABLE DOCUMENTS

2.1 General. See MIL-HDBK-240-1. This document can be found on ASSIST Online at https://assist.dla.mil.

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3. DEFINITIONS

3.1 Definitions. See MIL-HDBK-240-1.

4. PERSONNEL-BORNE ELECTROSTATIC DISCHARGE (PESD)

4.1 Introduction. This section addresses PESD safety and operability testing. It is the PESD test procedure for all conventional (non-nuclear) ordnance/munition types for the purpose of Joint service assessment. This procedure specifies PESD testing/analyses for all ordnance items including fielded bare energetic materials.

4.2 Scope.

4.2.1 Purpose. The purpose of PESD testing is to determine if exposure of ordnance items, ordnance electrical interfaces, and related systems, including but not limited to firing circuits and control electronics (weapon release, intent to launch, etc.) to the PESD environment test as defined in 4.3, due to personnel handling results in a safety and/or operability hazard. A safety hazard may be explosive material initiation, electrically initiated device (EID) initiation, defeating a safety-relevant function, or firing circuit activation, in accordance with MIL-STD-464 and MIL-STD-331. An operability hazard may be non-safety EID initiation, performance degradation, electrical interface malfunction, or dudding in accordance with MIL-STD-882, MIL-STD-464 and MIL-STD-331. Additionally, an operability failure may be Non-single Point Failure, Non-common-cause failure that may increase hazard risk. PESD exposure should be assessed or tested in the worst-case operational mode during the transportation/storage, assembly/disassembly, staged, handling/loading, and platform loaded S4 phases.

4.2.2 Application. Data obtained through use of this document should be included in the ordnance item/munition safety data package. PESD tests are applicable to all ordnance items (with and without EIDs), ordnance subsystems, ordnance related equipment, launchers, remote control devices, EIDs and firing/monitoring circuits, including those installed external to aircraft and vehicles.

PESD tests will be applicable at the equipment/system/AUR level and/or at the device level, depending on the configuration, assembly, and use of the item in the field. The PESD discharges should be applied to selected test points on the test item, without regard to procedural controls for the test item (such as the need to wear a static dissipative wrist strap and ground before handling the item), which could possibly be contacted by personnel during the different configurations/modes of operation during an S4 phase including: transportation/storage, assembly/disassembly, staged, handling/loading, and platform loaded. Unauthorized, non-standard configurations do not need to be addressed, such as disassembly of an AUR that is not authorized in the field. Specific PESD test fundamentals are identified in 4.3.

4.2.3 Limitations. This document is only applicable to conventional (non-nuclear) munitions to aid in the item, device, system, or AUR safety and operability assessment.

4.2.4 Implementation of Guidance. All items containing any energetic material or that control items containing energetic material should remain safe and without operational susceptibilities during and after exposure to the PESD test. All such items should meet their

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performance requirements during and after exposure to the PESD test environment, with parameters specified by MIL-STD-464.

4.2.5 PESD Guidance. All ordnance devices and AUR items should be PESD tested and/or assessed for in the Bare/man-carry configuration, that is, out of the packaged configuration with all protection devices removed, such as shorting clips, etc. It is recommended that other test configurations be considered when the item(s) are in their authorized configurations based on their S4 phases.

4.3 PESD Test Fundamentals.

4.3.1 Discharge Points. PESD testing should be performed to any point on the device/AUR that may be contacted by personnel or become charged during the transportation/ storage, assembly/disassembly, staged, handling/loading, and platform loaded configurations. Testing for each configuration should be in the worst-case operational mode (powered-on, unpowered, armed, etc.). Test points should include accessible pins on ordnance electrical interfaces, non-metal components, and related items, including but not limited to firing circuits and control electronics (weapon release, intent to launch, etc.). The assessment from 4.4 of this document should be used to determine the discharge points.

4.3.2 Bare Devices. Bare devices are ordnance components containing electrically initiated devices (EIDs) such as electrical initiators, exploding foil initiators, detonators, etc., in an AUR that have either one or both pins accessible on an external connector. These may include energetics such as explosives, or propellants or energetic initiating components such as: EID, Fuze Safe & Arm (S&A), Ignition Safety Devices. Bare Devices should be tested as per TABLE I using a stimulus circuit per FIGURE 1 electrical schematic. Additionally, these devices should be tested or evaluated per TABLE II as reduced voltages may expose a susceptibility.

All EIDs and all electro-explosive components that are not in a Faraday cage (as described in 4.5.1) should be tested according to TABLE I and the electrical schematic of FIGURE 1. The potential configuration for ESD discharge susceptibility falls into three distinct categories that require evaluation:

a. pin-to-pin, where the ESD pulse travels across the bridgewire, resulting in bridgewire heating

b. pin-to-case, where the case is isolated, the ESD discharge can occur through the energetic

c. shorted pins-to-case, where the ESD discharge can occur through the energetic

The shorted pins-to-case configuration exists during unpacking of the EID, most packaged EID's leads are shorted by the manufacturer. The shorted pin to case configuration can also be found in an AUR when shorting plugs or shorting relays are employed and leads are accessible via connector pins.

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EIDs should be tested explosively live in the pin-to-pin mode, the pin-to-case mode, and the shorted pins-to-case mode. This test methodology matches testing per MIL-DTL-23659F. After the PESD test, these devices must be subjected to the established no-fire and all-fire values to verify that they have not been sensitized or damaged as result of testing. Generally, an EID as qualified/tested per MIL-DTL-23659F need not be retested as part of the AUR as described in 4.5.2.

All percussion-initiated devices or components that are not in a Faraday cage (as described in 4.5.1) should be tested according to TABLE I and the electrical schematic of FIGURE 1. All percussion-initiated devices or components should be tested explosively live. Test designs should follow the precepts detailed in paragraph 4.4.

TABLE I. PESD Test Sequence for Bare Devices

Voltage (kV)	Capacitance (pF)	Discharge Inductance (µH)	Series Resistance (Ω)
+ 25 ± 2%	500 ± 5%	5 max	5000 ± 5%
− 25 ± 2%	500 ± 5%	5 max	5000 ± 5%
+ 25 ± 2%	500 ± 5%	5 max	500 ± 5%
− 25 ± 2%	500 ± 5%	5 max	500 ± 5%

NOTE: Use both a 500 ohm and 5000 ohm resistance to maintain consistency with MIL-DTL-23659F and MIL-STD-331 requirements. Each test sequence must consist of all four discharges. No additional resistance or inductance should be added to the circuit except for incidental inductance in the capacitor, resistor, and wiring.

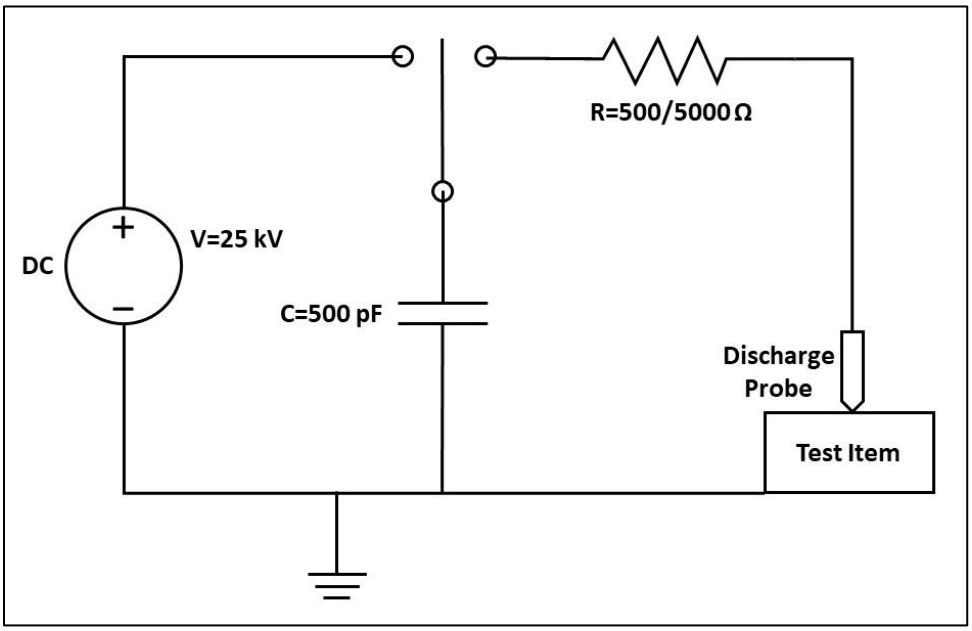


FIGURE 1. PESD electrical schematic

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TABLE II. ESD Test Sequences for Reduced/Incremental Voltage Tests

Voltage (kV)	Capacitance (pF)	Discharge Inductance (µH)	Series Resistance (Ω)
+ 5 ± 2%	500 ± 5%	5 max	500 ± 5%
− 5 ± 2%	500 ± 5%	5 max	500 ± 5%
+ 10 ± 2%	500 ± 5%	5 max	500 ± 5%
− 10 ± 2%	500 ± 5%	5 max	500 ± 5%
+ 15 ± 2%	500 ± 5%	5 max	500 ± 5%
− 15 ± 2%	500 ± 5%	5 max	500 ± 5%
+ 20 ± 2%	500 ± 5%	5 max	500 ± 5%
− 20 ± 2%	500 ± 5%	5 max	500 ± 5%
+ 25 ± 2%	500 ± 5%	5 max	500 ± 5%
− 25 ± 2%	500 ± 5%	5 max	500 ± 5%

NOTE: No inductance should be added to the circuit except for incidental inductance in the capacitor, resistor, and wiring. Each test sequence must consist of all ten discharges.

4.3.3 AUR Devices. AUR devices can be divided into two general categories for the assessment and testing of ESD:

4.3.3.1 AUR Devices with External Electrical Connectors to Energetic Components. These are AUR devices with electrical connectors that interface directly to EIDs or electro-explosive components or may be exposed to the PESD environment through arcing or significant impedance coupling. Additionally, these are AURs with PESD exposed energetic components, such as mortar-round propelling charges.

These AUR devices and items should be tested "live" according to TABLE I and the electrical schematic of FIGURE 1. Because these devices and items have the potential for injection of ESD into the EIDs or electro-initiated components (e.g., S&A Devices, fuze, guidance section), the full ESD test with both 500Ω and 5,000Ω should be conducted. For this testing, the electrical pins are not evaluated in the pin-to-pin and pins-to-case method as is done for bare devices. Rather the case of the item under test is grounded, and ESD is injected at each test point.

4.3.3.2 AUR Devices Without External Electrical Connectors to Energetic Components. These AUR devices and items have no EIDs or components connected to the exterior electrical connector nor directly exposed to the PESD environment. These are handled as a special case. Because there is no direct electrical connection to EIDs or electro-explosive component a reduced testing series may be utilized. This test is conducted to assess system operational susceptibilities of the AUR electronics. Testing of these devices should utilize TABLE III and the electrical schematic of FIGURE 1. Reduced voltage stimulus testing as per TABLE II is applicable for an AUR with electronic circuits, or EID where reduced or incremental voltages may expose a susceptibility that may not be possible at the 25 kV level due to ground discharge such as a dielectric breakdown path to ground or filtering characteristics.

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Requirements are different in the AUR configuration because the pins for the EIDs are not accessible and this test configuration is worst case. Each test sequence must consist of both positive and negative discharges.

TABLE III. PESD Test Sequences for AUR Devices and Operability

Voltage (kV)	Capacitance (pF)	Discharge Inductance (μH)	Series Resistance (Ω)
+ 25 ± 2%	500 ± 5%	5 max	500 ± 5%
− 25 ± 2%	500 ± 5%	5 max	500 ± 5%

NOTE: No inductance should be added to the circuit except for incidental inductance in the capacitor, resistor, and wiring.

4.3.4 Performance Requirements. During and after exposure to the PESD test, the device/AUR, including associated launchers, remote control devices, and associated electronics should meet the performance requirements detailed in their specifications with no safety or operability/performance failures or malfunctions, and without resetting or re-powering the device or AUR. Momentary upsets that auto-correct (without operator action) and have no effect on the operation of the item, such that it produces all required outputs at the specified time, are allowed.

4.4 PESD Pre-Test Assessment. Use of previous analyses and/or test data from previous variants of the same AUR may be evaluated and incorporated into the test planning and pre-test assessment in order to enhance, and/or reduce, the scope of the new testing, and to provide justification for the new test item configurations and test approach. A pre-test assessment is performed to include the following for test planning:

a. Determine PESD energy points of entry (points possibly contacted by personnel during assembly, handling, etc. or possible entry points from air or contact discharges) through an analysis of the complete drawing package that specifically identifies all the materials of construction for the device or AUR. ESD entry points include switches, displays, battery terminals, output terminals, electrical interfaces, device joints and discontinuities, and non-conductive surfaces, etc.

b. Determine the PESD external and internal current paths through an analysis of the electrical interface and mechanical interface drawings for the device or AUR. This may not be possible with non-metallic materials or with internal coupling paths.

c. Determine the most susceptible test item configurations and possible modes of operation based on personnel handling without regard to procedural controls during any of the S4 phases (bare, assembly/disassembly, handling/loading, powered on, armed, etc.).

d. Determine test item data required to be monitored or tested during or after PESD testing to evaluate the safety and operability/performance effects of the test item. This includes bridgewire resistance, communication links, programming modes, firing circuits, S&A indicators, etc. that are part of the device or AUR.

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e. Determine test item PESD protection features and their predicted performance, and whether they are repairable or replaceable.

f. Determine procedures and equipment required to verify the post-PESD test performance for each device, circuit, AUR, and non-Faraday cage component.

4.5 Previous Test Data and/or Analysis Requirements. The following are exceptions to conducting the PESD test on a new or modified ordnance device or AUR, or a subcomponent of the AUR. These exceptions only cover the energetic items, and any circuitry that is part of the AUR that was previously tested or analyzed that has not been changed as specified below. Any changes to the circuitry or energetic items must be addressed in the testing for the new/modified ordnance device or AUR, or a subcomponent of the new/modified AUR.

4.5.1 Faraday Cage Design. Any device, AUR, or subcomponent of an AUR, that is completely surrounded by metal with no electromagnetic entry points and has no external electrical connectors/leads or conductive mix primer/initiator is by design a Faraday cage and is safe for bare PESD exposure, provided all of the following are met:

a. Portions of the device designed to afford protection are constructed of metal material of minimum 0.4 mm (0.016 inch) thickness and with low bonding resistance (less than 1 ohm) across any non-welded sections. Bonding contact between ordnance sections should be uniformly distributed as to not allow significant difference of potential between bodies during the dynamic ESD event. Large potentials across bodies/sections may also develop due to impedance.

b. For all components surrounding the energetic material, drawings or technical documentation of the component materials, or specific delineation of the materials and configurations, by a competent authority should be provided for verification as an ESD Faraday cage. Official signed documentation is preferred. Performance specifications are not acceptable if the component materials are not specified for the device, AUR, or subcomponent of an AUR.

If the device or AUR does not meet all of the above, then verification should be accomplished via testing.

4.5.2 Other Qualifications. If the EID was tested bare and officially qualified in accordance with MIL-DTL-23659F additional PESD testing is not required. Documentation provided should demonstrate that the EID has been tested or qualified to MIL-DTL-23659F. Documentation should also be provided demonstrating that the EID in the AUR parts list is the same configuration as the EID that was tested or qualified per MIL-DTL-23659F. Electronic components such as filtering, switching or interlock devices connected to EIDs that have been tested or qualified per MIL-DTL-23659F should be tested for operability. It is important that interlocks that protect safety-critical functions such as ignition safety devices, or fuzes be evaluated for safety concerns as per the procedures and recommendations herein. JOTP-062 or MIL-STD-331 qualified safety interlocks meet this intent.

If the device, AUR, or a subcomponent of an AUR was tested live in accordance with the requirements of Allied Environmental Conditions and Test Procedures (AECTP) 508 Leaflet 2,

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and the official test report provides the test data required in this document to evaluate both safety and operability of the item, additional testing may not be required.

4.5.3 Previous Test Data. Previous test results and/or analysis on a variant may be used in the certification of a new item to reduce the scope of testing if the previous testing completely documented the safety and operability requirements for PESD as detailed in this document and if the following apply:

The new item/AUR is a variant of a previously fielded device, AUR, or subcomponent of an AUR. As a minimum, the new variant should be analyzed and tested for:

a. Changes to energetic components

b. Changes to system construction materials

c. Changes to electrical connectors/firing circuits

Only the new energetic items, sections with different materials of construction, or new electrical interfaces/firing circuits should be tested in accordance with the requirements of this document.

If the new item contains energetic devices that were PESD tested through previous testing in accordance with the requirements of this document, and no changes to the materials of construction for the item or AUR have been made, then these energetic devices do not need to be tested for the new item for safety concerns. However, for any new or modified firing circuits or other circuitry in the new AUR or new subcomponent of the AUR, these circuits should be tested for operability/performance, along with any other energetic devices that do not have live test results. Changes in the internal configuration of integrated circuit (IC) components such as Application Specific Integrated Circuits (ASIC), Field Programmable Gate Arrays (FPGA) and Programmable Logic Devices are considered modified circuitry and should be tested for operability.

4.6 Test Equipment and Instrumentation

4.6.1 Tolerances. Unless otherwise stated, test equipment accuracy/tolerances should meet the following minimum requirements:

Distance: ± 5 %
Frequency: ±2 %
Amplitude, ± 2 %
Time (waveforms): ± 2 %
Resistors: ± 5 %
Capacitors: ± 5 %
Current: ± 2 %
Voltage: ± 2 %

4.6.2 Discharge Equipment. The voltage, capacitance, resistance, and discharge circuit inductance, including the inductance of the capacitor and wiring to the probes should have the

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values of TABLE IV and tolerances listed in 4.6.1. Inductance should be measured at a nominal 1 kHz frequency.

TABLE IV. Discharge Equipment Characteristics

Voltage (kV)	Capacitance (pF)	Discharge Inductance (µH)	Series Resistance (Ω)
+ 25	500	5 max	5000
− 25	500	5 max	5000
+ 25	500	5 max	500
− 25	500	5 max	500

Note: Use both a 500 ohm and 5000 ohm resistance to maintain consistency with MIL-DTL-23659F and MIL-STD-331 requirements. No additional resistance or inductance should be added to the circuit except for incidental inductance in the capacitor, resistor, and wiring.

4.6.3 Discharge Methods. There are two acceptable methods for discharge that can be employed for PESD testing: direct contact discharge and approaching probe discharge.

a. Direct Contact Discharge: This method uses a discharge electrode from the test generator that is placed in direct contact with the selected test point, and the discharge is triggered (manually or remotely) with a discharge switch in the test generator. This is the test method that must be used for bare device testing and for pins on an electrical connector to maintain compatibility with MIL-DTL-23659F requirements.

b. Approaching Probe Discharge: This method positions the discharge electrode from the test generator approximately 1 foot from the selected test point, and the discharge electrode is moved at a minimum speed of 1 foot per second to the test point to initiate a discharge. If there is a possibility that the discharge could arc to another surface instead of the selected test point (such as a different pin on a connector), a salient (an electrical protrusion such as a wire) should be attached to the test point to direct the discharge to the selected discharge point without the possibility of arcing to an unintended discharge point.

In order to meet the PESD test methodology of this document, regardless of the discharge method used, the signal current monitored at the calibration test load must be within the limits identified in 4.6.4 of this document.

4.6.4 Discharge Equipment Calibration. The energy measured during calibration should be as defined in FIGURE 2 when using a 500 ohm series resistance, or FIGURE 3 when using a 5000 ohm series resistance of the energy stored on the capacitor. The upper and lower limits for the waveforms are based on the maximum and minimum energy output of the circuit in FIGURE 1 using the component tolerances in TABLE I - TABLE III. For additional information on these limits, see the PESD calculations in APPENDIX A for the 500 ohm circuit and in APPENDIX B for the 5000 ohm circuit.

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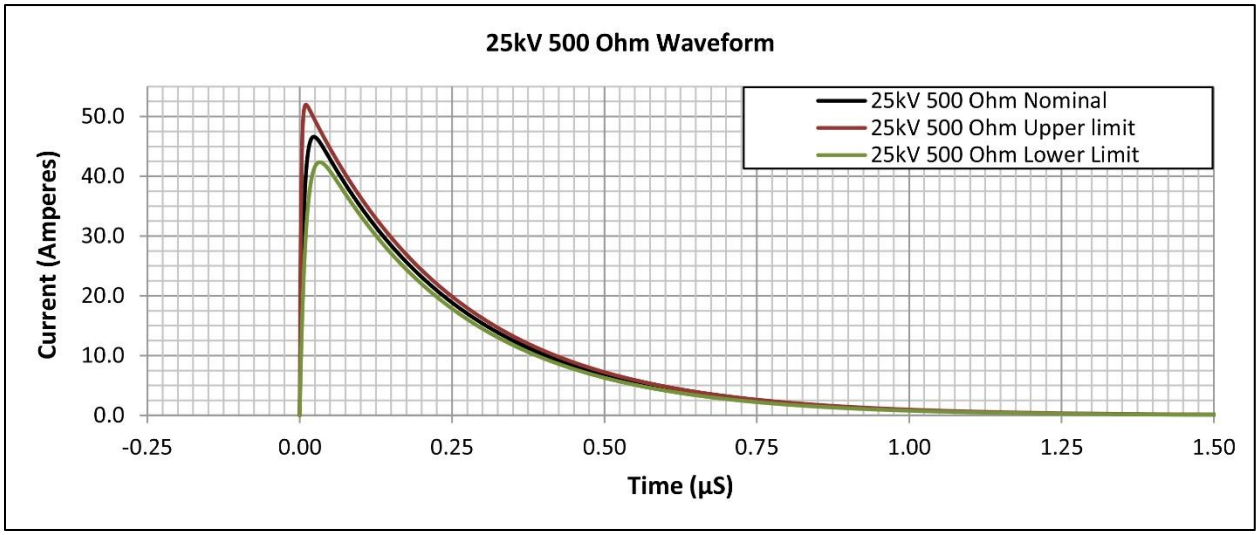


FIGURE 2. 500 ohm PESD current waveforms with limits

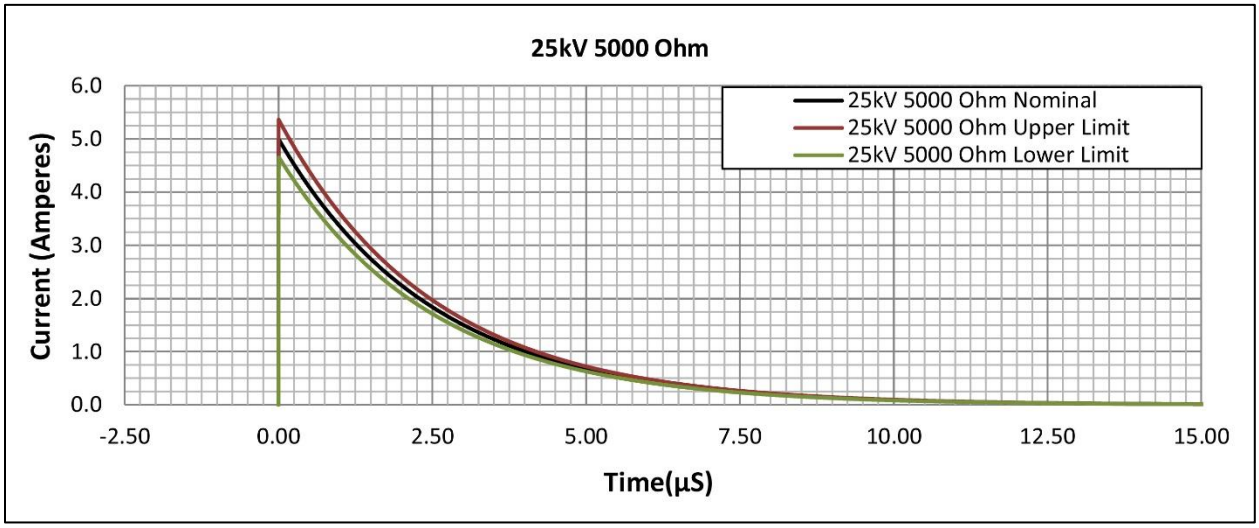


FIGURE 3. 5000 ohm PESD current waveforms with limits

A shorted discharge (for example, direct discharge to ground) to the ground plane from the test electrode using a current transformer on the output as shown in FIGURE 4 may be used to generate a current waveform with the upper and lower limits as shown in FIGURE 2 and FIGURE 3. The calculations for the current based waveforms can be found in APPENDIX A for the 500 ohm circuit and in APPENDIX B for the 5000 ohm circuit.

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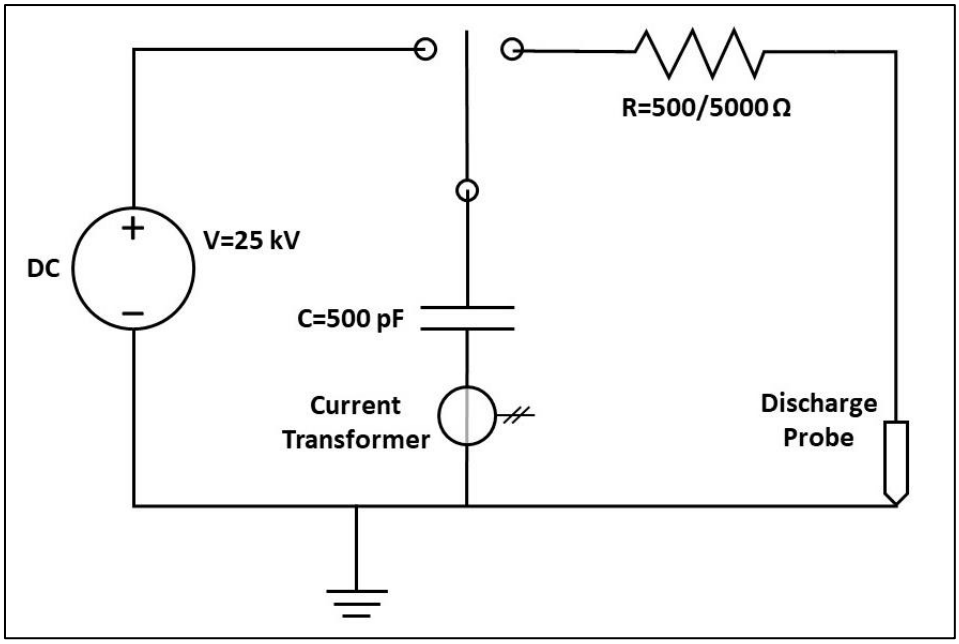


FIGURE 4. PESD shorted discharge current calibration

The output from the test equipment should be verified and recorded on a daily basis prior to testing and at the conclusion of testing for the day in accordance with the calibration limits of FIGURE 2 and FIGURE 3. If a salient is used on the test item, it should be considered part of the discharge circuit and should be used during calibration.

All measurement equipment must be calibrated for the test data to be valid. Equipment claiming MIL-STD-331 or MIL-STD-464 compliance does not guarantee the equipment actually meets the performance requirements of those documents and full compliance should be verified.

4.6.5 Instrumentation and Data Collection. In general, there are numerous valid methods to collect, digitize and analyze the pulse waveforms of interest. Before testing is conducted, an analysis of the instrumentation circuit should be conducted. Calculations should be made to determine if the data collection equipment and setup has the resolution to properly record the waveform of interest accurately. The useable risetime in the case of current transformers should be considered. The resistance of a voltage divider, commonly known as a "Pelligrini target" should be known and considered as a factor in the instrumentation circuit. In general, a factor of 10:1 higher bandwidth speed than the signal of interest should be used to gauge the suitability of the instrumentation. Further, any factors that affect the signal should be noted and summed to give an overall attenuation factor for the instrumentation circuit. Prior to any data collection, it should be fully understood what signal levels will be collected and digitized based on the expected current values for the discharge circuit. The attenuation factor should be noted and accounted for in the digitizer or included as a portion of the waveform analysis. It is especially important to be aware of the oscilloscope or digitizer input impedance. Modern instrumentation typically offers user switchable input impedances of 50Ω and 1MΩ. If the 50Ω input impedance is selected, a voltage reduction by a factor of 2:1 will occur for 50Ω impedance circuits. It is critical to understand this factor, and account for this in the analysis of

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the circuit attenuation. The following list includes equipment that is typically utilized for ESD data collection measurements:

a. Current and differential voltage probe

b. Oscilloscope

c. Digitizer

d. Power supply

e. Voltage divider

f. Data collection software and computer

Additional waveform filtering and smoothing functions should be avoided. These functions could be either in-circuit or digital. The functions that average the waveform, filter noise and statistically average can often mask underlying system issues that should be assessed and corrected. If an ESD generator is producing excessive noise and unwanted ringing, the generator circuit should be assessed rather than smoothing the waveform to coerce into specification.

It is critical to document all factors of the instrumentation circuit in the reporting or test notes. This will ensure that any person reading the report will have sufficient data to understand the data collected, and ensure it is within specifications.

4.6.6 Test Instrumentation. The test measurement equipment should have:

a. A minimum bandwidth of DC to 500 MHz.

b. A minimum rise time of 2 nanoseconds.

c. Digitizer Effective number of Bits should be a minimum of eight.

The measurement noise and jitter floor of the integrated test configuration should be verified through histograms or other means and sources of noise or errors are identified and controlled to benign levels.

Use a minimum digitizer sampling rate of three times the highest fundamental frequency to account for signal conditioning and Analog/Digital conversion.

4.6.7 Test Instrumentation Calibration Guidance. A description of how the Test Measurement and Diagnostic Equipment and test set-up meets the requirements of the Department of Defense Standard Practice for Calibration and Measurement Requirements per MIL-STD-1839 should be provided assuring system and equipment operational integrity and accuracy. Some of the items that should be addressed are:

a. Instrumentation calibration traceability as per MIL-STD-1839.

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b. Tests that will be conducted assuring that cabling, connector, calibration, pulse loads and interfaces test/verification through the use of network analyzer, phase-gain analyzer or equivalent are in the bandwidth of the measured ESD pulse.

c. Integrated instrumentation Test Uncertainty Ratio is addressed and confirmed to meet the minimum 4:1 requirement in MIL-STD-1839.

d. Digitizer Effective number of Bits should be a minimum of eight.

e. The measurement noise and jitter floor of the integrated test configuration should be verified through histograms or other means and sources of noise or errors are identified and controlled to insignificant levels.

f. Use a minimum digitizer sampling rate of three times the highest fundamental frequency to account for signal conditioning and Analog/Digital conversion.

4.7 Test Sample Quantities. The minimum sample size for AUR PESD testing should be 22 items for all categories of ordnance. No inert testing is allowed unless the AUR or the subcomponent of the AUR has been verified as a Faraday cage as defined in this document or has previously been tested live in accordance with 4.3 of this document. If fewer than 22 items are available for testing, the number of discharges delivered must be equivalent to 22 test sequences. This increases the possibility of failure due to additional discharges to the reduced sample set.

4.8 Test Sequences. A test sequence is defined as a complete series of discharges applied to the test points identified in the pre-test assessment (4.4 of this document). Each series of discharges is identified by TABLE I (4 total discharges per test point) and TABLE III (2 total discharges per test point), as applicable. Subsequent sequences may be conducted by using different items/munitions, or on the same item/munition with a different set of EIDs and electronic/electrical subsystems. If the subsequent testing uses the same device or AUR with the same EIDs, functional verification and bridgewire resistance testing should be performed after each series of discharges as defined above, unless documented and justified in the test plan. An equivalent of 22 test sequences should be conducted when testing to the PESD threat in order to demonstrate 90% operability with 90% confidence.

4.9 Incremental Voltage Testing. A minimum of four devices of the 22 samples required for PESD testing in the tested configuration (bare or AUR) must receive incremental voltage testing using the component requirements of TABLE II (one positive and one negative discharge through a 500 ohm resistor) and voltages of 5 kV, 10 kV, 15 kV, 20 kV, and 25 kV to document the safety and operability of the item to all possible PESD exposures. Even for the EIDs that receive incremental testing, all EIDs must receive both the + and – 25 kV discharge. Two of the devices should be tested using positive voltages with one device tested starting at 5 kV and increasing the voltage up to 25 kV in 5 kV steps (and must also include a – 25 kV discharge), and the other device starting at 25 kV and decreasing the voltage in 5 kV steps down to 5 kV (and must also include a – 25 kV discharge). The other two devices should be tested using negative voltages in a similar fashion (one device starting at – 5 kV and decreasing the voltage to – 25 kV in 5 kV steps (and must also include a +25 kV discharge), and the other device starting at – 25 kV and increasing the voltage in 5 kV steps up to – 5 kV (and must also

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include a +25 kV discharge)). Irrespective of the sample size used for the testing, the equivalent number of discharges delivered to the four samples must be delivered to the sample size used for testing (see TABLE V as an example). The four devices required for this testing should be from the 22 total samples required for testing.

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TABLE V. Example of Incremental Voltage PESD Testing On A Reduced Sample Set (4 sets of EIDs)

PESD Test Sequences 1-12		
Sequence No.[1]	**Test Set[2,3]**	**ESD Environment**
1	Set 1	+5 kV
1	Set 1	+10 kV
1	Set 1	+15 kV
1	Set 1	+20 kV
1	Set 1	+25 kV
1	Set 1	-25 kV
2	Set 1	-25 kV
2	Set 1	+25 kV
3	Set 1	+25 kV
3	Set 1	-25 kV
4	Set 1	-25 kV
4	Set 1	+25 kV
5	Set 1	+25 kV
5	Set 1	-25 kV
6	Set 1	-25 kV
6	Set 1	+25 kV
7	Set 2	+25 kV
7	Set 2	+20 kV
7	Set 2	+15 kV
7	Set 2	+10 kV
7	Set 2	+5 kV
7	Set 2	-25 kV
8	Set 2	-25 kV
8	Set 2	+25 kV
9	Set 2	+25 kV
9	Set 2	-25 kV
10	Set 2	-25 kV
10	Set 2	+25 kV
11	Set 2	+25 kV
11	Set 2	-25 kV
12	Set 2	-25 kV
12	Set 2	+25 kV

PESD Test Sequences 13-22		
Sequence No.[1]	**Test Set[2,3]**	**ESD Environment**
13	Set 3	-5 kV
13	Set 3	-10 kV
13	Set 3	-15 kV
13	Set 3	-20 kV
13	Set 3	-25 kV
13	Set 3	+25 kV
14	Set 3	+25 kV
14	Set 3	-25 kV
15	Set 3	-25 kV
15	Set 3	+25 kV
16	Set 3	+25 kV
16	Set 3	-25 kV
17	Set 3	-25 kV
17	Set 3	+25 kV
18	Set 4	-25 kV
18	Set 4	-20 kV
18	Set 4	-15 kV
18	Set 4	-10 kV
18	Set 4	-5 kV
18	Set 4	+25 kV
19	Set 4	+25 kV
19	Set 4	-25 kV
20	Set 4	-25 kV
20	Set 4	+25 kV
21	Set 4	+25 kV
21	Set 4	-25 kV
22	Set 4	-25 kV
22	Set 4	+25 kV

NOTES:

1. The Test Sequence includes all personnel contact points.
2. All electronic circuits/firing circuits and components must be verified to function normally prior to and after PESD testing of each set of EIDs.
3. Prior to and after the PESD test, bridgewire resistance measurements must be measured for each set of EIDs.

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4.10 Test Item Configuration. If the device or AUR contains electrically initiated devices and/or electrically initiated devices connected to an external connector that have not been tested to the requirements of this document or if MIL-DTL-23659F data is not available, then testing of the live EID(s) in the AUR is required. Testing of bare, live EIDs should be conducted for the pin-to-pin, pin-to-case, and shorted pin-to-case configurations. Testing of the device or AUR with live EIDs installed should be to each external electrical contact and non-metallic, or non ESD protected entry point. As a minimum, 22 samples are required to be tested using a direct contact discharge only.

a. EIDs must be tested with live EIDs in the AUR configuration for the following cases:

   1) if they are contained in a non-Faraday cage AUR and not connected to an exterior electrical connector, or

   2) if they have not previously been tested in the same AUR configuration, or

   3) if the AUR has different circuitry leading to the EID than a previously tested version.

b. Percussion initiated devices:

   1) Percussion initiated devices not in a documented Faraday cage or other non-electric devices must be tested live.

   2) Test configuration should be a bare device (not installed) if the item will be used in a standalone configuration or in the AUR if the item will only be in the AUR configuration.

4.11 Configuration Changes. Any change to a previously PESD tested and/or certified device or AUR, including changes to component materials, energetic materials, EIDs, packaging, electrical interfaces, communication interfaces, or remote devices should be subjected to PESD testing to verify the new configuration meets the PESD requirements of this document. A new nomenclature/cataloging number should be assigned.

4.12 Instrumented Testing. Instrumented testing should not be performed on electrically initiated devices that have not been tested live either at the device level or in the AUR configuration as defined in 4.3, until a Joint Service agreement on instrumentation methods and procedures are developed. Additional investigation is needed to develop a Joint Service instrumented testing methodology that will provide the safety and operability data for an EID; this methodology will include the specific instrumentation, data acquisition, and safety and operability margins for pin-to-pin, pin-to-case, shorted pins-to-case, and AUR PESD testing. Instrumented testing is currently only allowed for items where the electrical/firing interface has changed, but no changes have been made to the EID.

4.13 PESD Test Procedure Requirements.

4.13.1 Bare EID Discharges. Only the direct contact discharge method should be used for discharges to pairs of pins or leads in all combinations and between the shorted pins or leads

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(all pins or leads shorted to each other external to the device) and the case of the initiator. The pin-to-case and shorted pins-to-case discharge connection should be maintained for 60 seconds. Salients are to be used to direct the discharge to the pins or leads in the electrical connector.

4.13.2 Bare and AUR Device Discharges. Discharges should be made to all electrical interfaces using the direct contact discharge method including each pin on an external connector, control interface, or any other device that has an ordnance related function such as antennas or inductive antennas for fuzes. All plastic and non-metallic components of a device or AUR that contain ordnance, circuitry that controls the ordnance, or provides a path to the internal components of the device should receive discharges. Any other possible entry points including joints between sections and covers over cabling should also be tested. Salients must be used to direct the discharge to test points where arcing could occur to adjacent surfaces such as electrical pins on a connector. EIDs must be subjected to bridgewire resistance test and no-fire and all-fire stimulus after the PESD test to verify conformance with safety and performance specifications of the device. Bridgewire resistance measurements by themselves may not accurately assess the integrity of the EID.

4.13.3 Remote Control Item Testing. Items with a remote control should be tested as individual components (transmitter and receiver), and as a system with both components operating to determine if a discharge to one of the system components can cause a safety or operability consequence in the other component.

4.13.4 Arming. Operational conditions may require that an armed item be handled; therefore, PESD testing should include testing in the armed condition for those devices without regard to operational procedures.

4.13.5 Bridgewire Resistance Measurements. Prior to and after the PESD test, bridgewire resistance measurements must be made for all live EIDs in the device or AUR with a bridgewire measurement device for the specific devices being measured.

4.13.6 Verification of Circuit Functionality. All electronic circuits/firing circuits and components should be verified that they function normally prior to and after PESD testing.

4.13.7 Temperature and Humidity. Testing should be conducted on devices at an ambient temperature of +23°C±10°C (+73°F±18°F). The relative humidity of the test chamber should be no greater than 50 percent. The device should be preconditioned at +23°C±3°C (+73°F± 5°F), with a relative humidity no greater than 50 percent for no less than 24 hours prior to testing. If the size of the AUR precludes indoor controlled testing, testing should be performed at the lowest humidity possible, but as a minimum, there should be no moisture on the test item (non-condensing humidity level). It should be noted that ESD effects on ordnance at a low temperature/humidity may be different from the standard test temperature/humidity. APPENDIX C explains this phenomenon.

4.14 Post PESD Test Requirements.

4.14.1 Bridgewire Resistance. All EIDs subjected to live PESD testing should be measured for bridgewire resistance after PESD testing. The measured value should be used to

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verify that the resistance value is still within tolerance for the device specification. Any large changes in value or out of specification findings will be noted in the test report.

4.14.2 No-Fire and All-Fire Verification. All EIDs subjected to live PESD testing should be evaluated to ensure they still meet the specified device No-Fire and All-Fire parameters. The verification testing is intended to evaluate if the device will not fire when subjected to the specified No-Fire stimulus and will produce its expected output when subjected to the specified All-Fire stimulus. This validates that the device has not dudded and has not been sensitized as a result of PESD testing.

4.14.3 Post-test Inspection. All EIDs and percussion-initiated devices not in a documented Faraday cage or other non-electric devices subjected to live PESD testing will be tested for functionality after PESD exposure to verify proper operation.

4.14.4 Operability. A complete checkout/functional test of the device or AUR to include all electrical interfaces should be completed after each set of discharges to verify the item is functional and was not damaged or affected by the PESD test discharges; this includes initiation or dudding.

Items with a remote control should be verified to function normally after each set of discharges. Both transmit and receive functions should be tested and verified to be operational.

Any functional or operational anomaly that prevents the device or AUR from completing its function when commanded after the selected number of PESD discharges should be considered an operability failure.

4.14.5 Data Collection. Record the following for each test:

a. Stimulus waveform parameter data/plots.

b. Test configuration/test mode.

c. Photo or diagram of the test item showing discharge points.

d. Post-discharge functional check pass/fail results.

e. Visual damage inspection results.

f. Test item responses (e.g., components affected, visible reactions, burning, upsets, recovery times, etc.).

g. EID bridgewire resistance and determination of degradation, if required.

h. Date, time, location, temperature, and humidity for each discharge.

i. Any attenuation used in the measurement circuit.

j. Any termination loads used in the measurement circuit.

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4.15 PESD Test Assessment and Analysis Reports.

4.15.1 Data Sharing. In order to facilitate data sharing across the Services and to ensure test data consistency and consistent operational guidance, an ESD test report should be developed as part of every ESD test. All ESD test reports should be available for inclusion in a centralized DoD database.

4.15.2 Test Report. Test reports should contain the following information and documentation as a minimum:

4.15.2.1 Drawing Package A complete drawing package for the device or AUR identifying all materials of construction and all electrical interfaces should be provided.

4.15.2.2 PESD Test Record. The test record consisting of the following:

a. Date(s) of test

b. Test location/facility

c. Test personnel names

d. Test and conditioning records documenting compliance with the following:

1) All test equipment/instrumentation used to measure the waveforms, type/model/serial number, equipment/instrumentation calibration data, and including the calibration waveforms as specified in 4.6.4.

2) Configuration of test item, Mark (MK)/MOD, nomenclature/Department of Defense Identification Code (DODIC), model, drawing, part number, revision, National Stock Number (NSN), as applicable. The description should include any variations of the test item from the final configuration of the live ordnance item. The configuration should also include a complete description of the test item for PESD testing including if it was powered on, armed, electrical interface pins designated for each test, etc.

3) Test item record: Each test article should be identified and described by a test item record. The record should include pre-test performance, performance during the test, and post-test performance information and documents as described below:

4.15.2.2.1 Pre-test Performance. The operation/function of the device or AUR should be verified and documented prior to PESD testing to provide a reference for the operability assessment. This includes any functional tests required to verify proper operation of the device or AUR, including operation of remote-controlled devices. A record should be made of all data to determine compliance with required performance and, when applicable, to provide a reference level or criteria for checking desired performance of the test item during or at the conclusion of the test. If several tests are to be performed in sequence and the cumulative effect is desired, then the measurement of performance level prior to each individual test need not be conducted,

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and only the pre-sequence measurement performed. The pre-test performance check may be made after installation of the item under test if installation conditions necessitate it. Pre-PESD test bridgewire resistance measurements are part of the pre-test performance data as specified in 4.13.5.

4.15.2.2.2 Performance During Test. When operation of the test item is required during the test, a record should be kept of the data for comparison with pre-test or post-test performance as required. The conditions during the performance check should be those specified in the individual test.

4.15.2.2.3 Post-test Performance. The operation of the test item should be verified and documented at the conclusion of the test as specified in 4.3.4 and 4.14.4, and compared with pre-test performance or during-test performance, whichever is required for determining the operability of the item from the PESD testing. This functional test should include the operation and status of any remote-controlled device(s). Also, post PESD test bridgewire resistance measurements, no-fire test results, and all-fire test results are part of the post-test performance documentation that must be included.

4.15.2.3 Photographs. Photograph each test configuration and test point along with any salients used to direct PESD discharges to the specific test points.

4.15.2.4 Data. Record all test results, data, and ESD calibration waveforms including reduced voltage test results.

4.15.2.5 Analysis of Test Results and Data. Test data should be compared to and evaluated against the criteria set forth in MIL-STD-464, MIL- STD-331, the test item specification, and test item test plan to determine if the item met the required criteria. Deficiencies should be identified and documented. Safety and operability failures are evaluated separately. No safety failures are allowed for service issued items.

4.15.3 Conclusion of the ESD Sensitivity of the Item and Criteria for Assigning the PESD Code. FIGURE 5 provides a sample of the ESD Summary datasheet.

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ESD Summary Datasheet			
NALC/DODIC/CRC:	NSN:	MILITARY BRANCH: Choose One	P/N:
ITEM NAME:			DRAWING NUMBER:
			TEST ACTIVITY REFERENCE:
NOMENCLATURE:			TEST COMMAND:
			TEST REFERENCE DATE:
			SECURITY CLASSIFICATION: (Please note that this also needs to be changed at top and bottom of the page if this datasheet is classified.) Choose an item.
PESD CODE: Choose an item.	HESD CODE: Choose an item.		EVALUATION STANDARD: Choose an item.
CONTAINER TYPE:	CONTAINER MATERIAL:		
COMMENTS:			

FIGURE 5. Sample ESD Summary Datasheet

4.15.3.1 PESD Safe. In accordance with 4.2.4, no ordnance item may be fired. Ordnance use includes installation, firing or release of the ordnance. This requirement includes passing the reduced voltage tests.

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4.15.3.2 Operable. In accordance with 4.2.4, when the ordnance is provided its required inputs, it must perform to completion of its function and sequence producing all required outputs within the operating period or at the specified time after all PESD exposure. This requirement includes passing the reduced voltage tests also.

4.15.3.3 Susceptible. In accordance with 4.2.4, if the test item has remained safe, however has reduced operational capability related to ESD exposure it is considered susceptible to the environment.

4.15.3.4 Assignment of Joint Service PESD Code. At the conclusion of all testing and after analysis of all results, the test report should include a PESD Code as delineated in TABLE VI:

TABLE VI. PESD Code Descriptions

Code	Description
PS	Safe and operable. This indicates that the ordnance is both safe and operable for PESD exposure. It has passed live PESD testing for all EIDs and/or energetic material with no dudding or initiation, and has passed PESD testing to all electrical interfaces that control or interface to the item with no effect on performance or functioning as described in FIGURE 5 and 4.15.3.2 above.
PO	ESD Ordnance Susceptibility. This indicates that the ordnance has been evaluated or live PESD testing to all EIDs and energetic material with no safety-significant, safety functionality or hazard impact failures. The Ordnance Item can be expected to function through safe separation or safe escape, with minor deviations, however significant deviations may be present during mission completion. The item may not perform or function as designed. The operability failures can be expected to not attribute to the hazard risk of the item; major energetic components can be expected to not be initiated.
PH	ESD Ordnance Not Safe/Not Operable. This indicates that the ordnance failed live PESD testing or was not tested and/or evaluated to one or more EIDs or energetic material with either dudding, initiation, or both for one or more operable requirements. Exposure to the PESD environment has left the ordnance item in a condition: a) where it is armed b) that qualifies as a hazard risk c) where it cannot be expected to perform or function through safe separation or safe escape d) or that the ordnance item is completely non-responsive or in an unknown condition. This code is not intended for general issue, such as bare energetic components being shipped from a manufacturer to an assembly facility.

4.15.4 Safety Data Package. The results of all PESD testing and analysis conducted relevant to assessing the ordnance item/munitions safety for service should be compiled into a safety data package for review. The package must include the previously approved test plan, including the rationale for any variance from the joint requirements. In addition, any deviation

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from that approved plan should be presented along with an analysis showing why the results should be accepted. The package must also provide any safety and operability deductions derived from those results.

5. HELICOPTER-BORNE ELECTROSTATIC DISCHARGE (HESD)

5.1 Introduction. This section addresses HESD safety and operability testing. It is the HESD test procedure for all conventional (non-nuclear) ordnance/munition types for the purpose of Joint service assessment and testing. This procedure specifies HESD requirements for all ordnance items including bare energetic materials.

5.2 Scope.

5.2.1 Purpose. The purpose of HESD testing is to determine if exposure of ordnance items, ordnance electrical interfaces, and related systems, including but not limited to firing circuits and control electronics (weapon release, intent to launch, etc.) results in a safety and/or operational performance hazard when exposed to the HESD environment; this environment is created by helicopter and/or rotary-winged aircraft operations. These hazards may be ordnance initiation, EID initiation, firing circuit initiation, performance degradation, electrical interface malfunction, or dudding. HESD exposure should be assessed or tested for man-carry (bare, outside the shipping container), Vertical replenishment (VERTREP)/sling load, external carry, and hot tube loading/unloading.

5.2.2 Application. Data obtained through use of this document should be included in the ordnance item/munition safety data package. The HESD test requirements are applicable to all ordnance items (with and without EIDs), ordnance subsystems, ordnance related equipment, launchers, remote control devices, EIDs and firing/monitoring circuits, including those installed external to aircraft and vehicles, those carried outside the shipping container internal to the aircraft, and for hot tube loading operations.

HESD test requirements may be applicable at either or both the equipment/system/AUR level and the device level depending on the configuration, assembly, and use of the item in the field. The HESD requirements should be applied to selected test points on the test item, without regard to procedural controls for the test item (such as a requirement to wear a static dissipative wrist strap and ground before handling the item). These test points are defined as areas which could possibly make contact with a charged aircraft or become charged during the different configurations/modes of operation including: transportation/VERTREP/sling load, man-carry (internal, outside the shipping container), external carry, and hot tube loading/ unloading. Unauthorized non-standard configurations do not need to be addressed such as disassembly of an AUR that is not authorized in the field. Specific HESD test requirements are identified by the category of ordnance in 5.3.

5.2.3 Limitations. This document is only applicable to conventional (non-nuclear) munitions to aid in the item, device, system, or AUR safety and operability assessment. This document is not intended to be used to aid the assessment of effectiveness, reliability or performance of an ordnance item/AUR.

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5.2.4 Implementation of Guidance. All items containing any energetic material or that control energetic, should remain safe and operable, and will meet their performance requirements, during and after exposure to the HESD test requirements, with parameters specified by MIL-STD-464 and MIL-STD-331.

5.2.5 HESD Guidance. HESD should be tested and/or assessed for all ordnance devices and AUR items in the following exposures:

a. VERTREP/sling load for packaged items in the documented packaged configuration. This requirement applies to all items.

b. Bare/man-carry for items out of the packaged configuration and carried into or out of operating aircraft.

c. External carry for items mechanically and electrically attached to the exterior of aircraft and exposed to in-flight charging.

d. Hot tube loaded for items that are mechanically and electrically installed and connected on ungrounded operating aircraft.

5.3 HESD Requirements.

5.3.1 HESD Test Requirements. HESD testing should be to any point on the item/AUR that may be contacted or become charged during man-carry/internal transport, hot tube loading/unloading, external carry, and/or VERTREP/sling load, and in the appropriate worst case operational mode (powered on, unpowered, armed, etc.) in each configuration. This should include ordnance electrical interfaces (including but not limited to firing circuits, control electronics, weapon release, intent to launch, etc.). The assessment from 5.4 of this document should be used to determine the discharge points.

5.3.2 Bare/Man-Carry Requirements. Any device containing energetic material, EIDs, and/or percussion-initiated devices that are not in a Faraday cage (as defined in 4.5.1 and in 5.5.1) that could possibly be carried bare (outside the shipping container) onto, or off, a rotary wing aircraft will be tested according to the HESD requirements in TABLE VII. All items must be evaluated using live EIDs or energetic material if not in a Faraday cage in the man-carry (bare configuration) according to the electrical schematic of FIGURE 6. This requirement should apply to all items in the following commodity categories: Small Caliber Ammunition; Non-Lethal Ammunition; Man-Emplaced Ammunition; Man-Carried Missiles/Rockets; Pyrotechnics; Fuzes and Ignition Systems; and any air launched rockets/missiles that could be carried bare on/off an aircraft, and other items from the remaining commodity categories that could be carried bare on/off an aircraft.

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TABLE VII. HESD Requirements

Voltage (kV)	Capacitance (pF)	Discharge Inductance	Series Resistance (Ω)
+ 50 ± 2%	1000 ± 5%	< 20	< 1
- 50 ± 2%	1000 ± 5%	< 20	< 1
+ 100 ± 2%	1000 ± 5%	< 20	< 1
-100 ± 2%	1000 ± 5%	< 20	< 1
+ 150 ± 2%	1000 ± 5%	< 20	< 1
- 150 ± 2%	1000 ± 5%	< 20	< 1
+ 200 ± 2%	1000 ± 5%	< 20	< 1
- 200 ± 2%	1000 ± 5%	< 20	< 1
+ 250 ± 2%	1000 ± 5%	< 20	< 1
- 250 ± 2%	1000 ± 5%	< 20	< 1
+ 300 ± 2%	1000 ± 5%	< 20	< 1
- 300 ± 2%	1000 ± 5%	< 20	< 1

NOTE: No additional resistance or inductance should be added to the circuit except for incidental inductance in the capacitor, resistor, switching, and wiring.

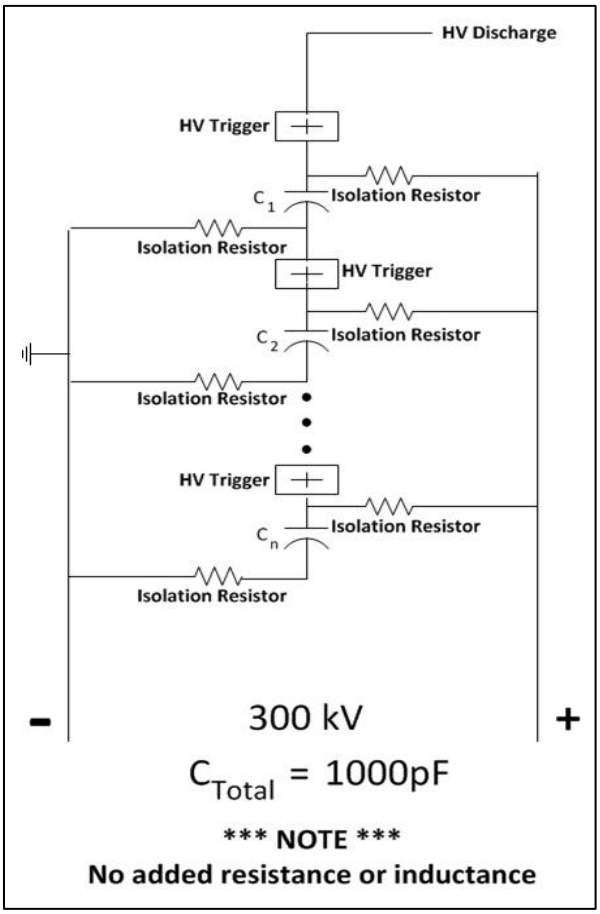


FIGURE 6. Bare/man-carry/hot tube loaded electrical schematic (Marx Generator Type)

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5.3.3 Hot Tube Loaded. Any device that contains energetic material, electrically initiated devices, and/or percussion initiated devices that are not in a Faraday cage (as defined in 4.5.1 and 5.5.1) that could possibly be hot tube loaded/unloaded into/out of a launcher, or onto an operating aircraft will be tested according to the HESD requirements in TABLE VII. All items must be evaluated using live EIDs or energetic material (if not in a Faraday cage) in the hot tube loaded configuration according to the electrical schematic of FIGURE 6.

5.3.4 External Carry. Any device that contains energetic material, electrically initiated devices, and/or percussion initiated devices that are not in a Faraday cage (as defined in 4.5.1 and 5.5.1) that could possibly be carried externally (outside the shipping container) on an aircraft will be tested according to the HESD requirements in TABLE VII. All items must be evaluated using live EIDs or energetic material (if not in a Faraday cage) in the external carry configuration according to the electrical schematic of FIGURE 7.

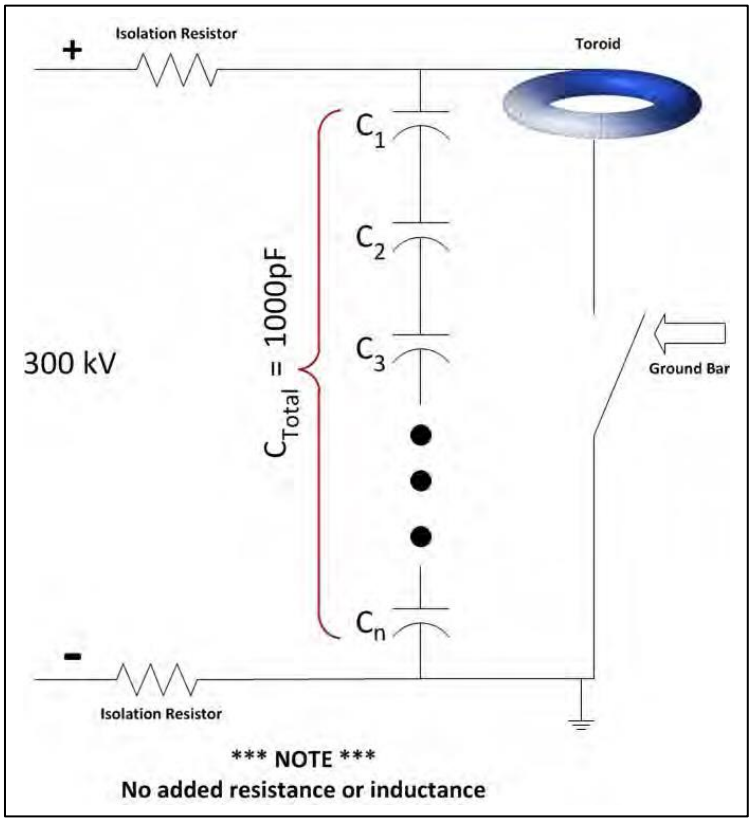


FIGURE 7. VERTREP/sling load/external carry electrical schematic

5.3.5 VERTREP/Sling Load Requirements. All devices that contain energetic material, electrically initiated devices, and/or percussion-initiated devices that are not in a Faraday cage package (as defined in paragraph 4.5.1 and 5.5.1) should be tested in the packaged VERTREP/sling load configuration according to the HESD requirements in TABLE VII. If the device/item contains only energetic material (percussion devices, no EID) that is not in a Faraday cage as defined in 4.5.1, then the VERTREP/sling load testing must include the live energetic material. All items from every ordnance category must be evaluated using live EIDs or energetic material according to the electrical schematic of FIGURE 7.

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5.3.6 Performance Requirements. During and after exposure to the HESD test environments, the energetic item or AUR, including associated launchers, remote control devices, and associated electronics should meet the performance requirements detailed in their specifications with no safety or operability/performance failures or malfunctions and without resetting or re-powering the device or AUR. Momentary upsets that auto-correct (without operator action) and have no effect on the operation of the item, such that it produces all required outputs at the specified time, are allowed.

5.4 Pre-HESD Test Assessment. Use of previous analyses and/or test data from previous variants of the same AUR may be evaluated and incorporated into the test planning and pre-test assessment in order to enhance and/or reduce the scope of the new testing, and to provide justification for the new item test configurations and test approach. A pre-test assessment is performed to include the following for test planning:

a. Determine HESD energy points of entry (points possibly contacted with a charged aircraft or entry points from air gap or direct contact discharges) through an analysis of the complete drawing package that specifically identifies all the materials of construction for the device or AUR, and the shipping/packaging configuration. Energy entry points include switches, displays, battery terminals, output terminals, electrical interfaces, containers/container joints or discontinuities, and non-conductive surfaces, etc.

b. Determine the HESD external and internal current paths through an analysis of the container, as well as the electrical and mechanical interface drawings for the device or AUR. This may not be possible with non-metallic materials or with internal coupling paths.

c. Determine the most susceptible test item configurations and possible modes of operation based on aircraft exposure without regard to procedural controls during transportation/VERTREP/sling load, man-carry transporting (on/off the platform), external platform carry, and hot tube loading/unloading.

d. Determine test item data required to be monitored or tested during or after HESD testing to evaluate the safety and operability/performance effects on the test item. This includes bridgewire resistance, communication links, programming modes, firing circuits, S&A indicators, etc. that are part of the device or AUR.

e. Determine test item HESD protection features, their predicted performance, and whether they are repairable or replaceable.

f. Determine procedures and equipment required to verify the post-HESD test performance for each device, circuit, AUR, and non-Faraday cage component.

5.5 Previous Test Data and/or Analysis Requirements. The following are exceptions to conducting the HESD test on a new or modified ordnance device or AUR, or a subcomponent of the AUR. These exceptions only cover the energetic items, and any circuitry that is part of the AUR that was previously tested or analyzed that has not been changed as specified below. Any

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changes to the circuitry or energetic items must be addressed in the testing for the new/modified ordnance device or AUR, or a subcomponent of the new/modified AUR.

5.5.1 Faraday Cage Design. Any device, AUR, or subcomponent of an AUR, that is completely surrounded by metal with no electromagnetic entry points and has no external electrical connectors/leads or conductive mix primer/initiator is by design a Faraday cage and is safe for bare HESD exposure, provided all of the following are met:

a. Portions of the device designed to afford protection are constructed of metal material of minimum 0.4 mm (0.016 inch) thickness and with low bonding resistance (less than 1 ohm) across any non-welded sections. Bonding contact between ordnance sections should be uniformly distributed as to not allow significant difference of potential between bodies during the dynamic ESD event. Large potentials across bodies/sections may also develop due to impedance.

b. For all components surrounding the energetic material, drawings or technical documentation of the component materials or specific delineation of the materials and configurations by a competent authority should be provided for verification as an ESD Faraday cage. Official signed documentation is preferred. Performance specifications are not acceptable if the component materials are not specified for the device, AUR, or subcomponent of an AUR.

If the device or AUR does not meet all of the above, then verification should be accomplished via testing.

5.5.2 Shipping Containers. Any shipping/packaging container that is completely made of metal and has no external electrical connectors/leads or openings (such as vents, etc.) is by design a Faraday cage/shield and provides protection to any device contained inside from HESD exposure, provided all of the following are met:

a. All components of the shipping/packaging container are constructed of metal material of minimum 0.4 mm (0.016 inch) thickness and with low bonding resistance (less than 1 ohm) across any non-welded components.

b. Official signed drawings documenting the materials of the shipping/packaging container are required to prove Faraday shield ESD designation. Performance specifications are not acceptable if the materials are not specified for the shipping/packaging container.

5.5.3 Shipping Container/Packaging Requiring Testing. If the device or shipping container/packaging does not meet all the requirements above, then testing is required.

The container/packaging containing the device or AUR with live EIDs and/or energetic material must be tested to the requirements of TABLE VII, using the electrical schematic of either FIGURE 6 or FIGURE 7, with 10 sets of positive and negative discharges to the thinnest and/or non-metallic section(s) of the container without melt/burn through, or initiation/dudding/ performance degradation of the item in the container.

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5.5.4 Previous Test Data. Previous test results and/or analysis on a previous variant may be used in the certification of a new item to reduce the scope of testing if the previous testing completely documented the safety and operability requirements for HESD as detailed in this document and if the following apply:

a. The new item/AUR is a variant of a previously fielded device, AUR, subcomponent of an AUR, or shipping container. The new variant and/or shipping container should be analyzed and tested for:

   1) Changes to energetic components

   2) Changes to system construction materials

   3) Changes to electrical connectors/electrical interfaces as a minimum

b. Only the new energetic items, sections with different materials of construction, or new electrical interfaces/firing circuits should be tested in accordance with the requirements of this document.

5.5.5 New Items. If the new item contains energetic devices that were HESD tested through previous live testing in accordance with this document, and no changes to the materials of construction for the item or AUR have been made then these energetic devices do not need to be tested for the new item for safety concerns. However, for any new or modified firing circuits or other circuitry in the new AUR, or new subcomponent of the AUR, for previously HESD tested items, those circuits should be tested for HESD vulnerability, along with any other energetic devices that do not have live test results.

5.6 Test Equipment and Instrumentation.

5.6.1 Tolerances. Unless otherwise stated, test equipment accuracy/tolerances should meet the following minimum requirements:

Distance: ± 5 %
Frequency: ±2 %
Amplitude: ± 2 %
Time (waveforms): ± 2 %
Resistance: ± 1 %
Capacitors: ± 5 %
Current: ± 2 %
Voltage: ± 2 %

5.6.2 Discharge Equipment. The voltage, capacitance, resistance, and discharge circuit inductance, including the inductance of the capacitor and wiring to the probes should be in accordance with the values in TABLE VII. Inductance should be measured at a nominal 1 kHz frequency.

5.6.3 Discharge Methods. There are two acceptable methods for discharge that can be employed for HESD testing, direct contact discharge and approaching probe discharge:

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a. Direct Contact Discharge: This method uses a discharge electrode from the test generator that is placed in direct contact with the selected test point and the discharge is triggered (manually or remotely) with a discharge switch in the test generator. This is the discharge method that must be used for man-carry/bare and hot tube loading test configurations and conducted using the electrical schematic of FIGURE 6.

b. Approaching Probe Discharge: This method positions the test generator's discharge electrode approximately 4 feet from the selected test point and the discharge electrode is moved at a minimum speed of 2 feet per second to the test point to initiate a discharge. If there is a possibility that the discharge could arc to another surface instead of the selected test point (such as a different pin on a connector), a salient will be attached to the test point to direct the discharge to the selected discharge point without the possibility of arcing to an unintended point. This is the discharge method that must be used for VERTREP/sling load and external carry test configurations and conducted using the electrical schematic of FIGURE 7.

In order to meet the HESD requirements of this document, regardless of the discharge method used, the energy measured during the calibration discharge must be within the calibration limits of section 5.7.

5.6.4 Circuit Schematics. FIGURE 6 defines a sample circuit schematic to be used for bare/man-carry and hot tube loaded HESD testing. FIGURE 7 defines the circuit schematic to be used for VERTREP/sling load and External carry HESD testing. It is of note that there are different configurations for each circuit that can be implemented and will produce the required output. An exact duplicate of the circuits shown in FIGURE 6 and FIGURE 7 is not necessarily required.

5.6.5 Instrumentation and Data Collection. In general, there are numerous valid methods to collect, digitize and analyze the pulse waveforms of interest. There are numerous areas to consider as well as data to report to yield consistent, repeatable and readable results. Before testing is conducted, an analysis of the instrumentation circuit should be conducted. Calculations should be made to determine if the data collection equipment and setup has the resolution to properly record the waveform of interest. The useable risetime in the case of current transformers should be considered. In general, a factor of 10:1 higher bandwidth speed than the signal of interest should be used to gauge the suitability of the instrumentation. Further, any factors that affect the signal should be noted and summed to give an overall attenuation factor for the instrumentation circuit. Prior to any data collection, it should be fully understood what signal levels will be collected and digitized based on the expected current values for the discharge circuit. The attenuation factor should be noted and accounted for in the digitizer or included as a portion of the waveform analysis. It is especially important to be aware of the oscilloscope or digitizer input impedance. Modern instrumentation typically offers user switchable input impedances of 50Ω and 1MΩ. If the 50Ω input impedance is selected, a voltage reduction by a factor of 2:1 will occur for 50Ω impedance circuits. It is critical to understand this factor, and account for this in the analysis of the circuit attenuation.

Additional waveform filtering and smoothing functions should be avoided. These functions could be either in circuit or digital. Functions that average the waveform, filter noise and

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statically average can often mask underlying system issues that should be assessed and corrected. If an ESD generator is producing excessive noise and unwanted ringing the generator circuit should be assessed rather than smoothing the waveform to coerce into specification.

It is critical to document all factors of the instrumentation circuit in the reporting or test notes. This will ensure that any person reading the report will have sufficient data to understand the data collected, and ensure it is within specifications.

5.6.6 Test Instrumentation. In order to verify the HESD waveforms, the measurement equipment should have:

a. Minimum bandwidth of DC to 500 MHz

b. Minimum rise time of 20 nanoseconds

5.7 Calibration Requirements. The output measured during calibration should be as defined in FIGURE 8 through FIGURE 10. The upper (i_{UL}) and lower (i_{LL}) limits for the waveforms are based on the maximum and minimum output of the circuits in FIGURE 6 and FIGURE 7 using the component tolerances in 5.6.1. Calibration test waveforms should fall within the upper (FIGURE 8) and lower (FIGURE 10) limits. For additional information on these limits, see APPENDIX D.

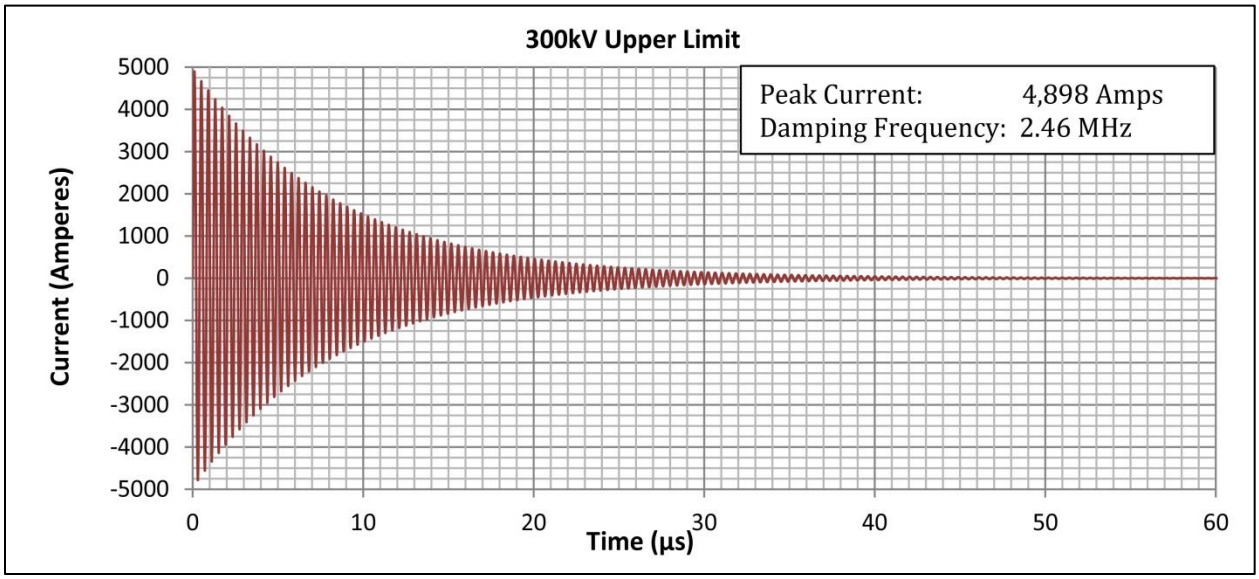


FIGURE 8. HESD upper limit current waveform

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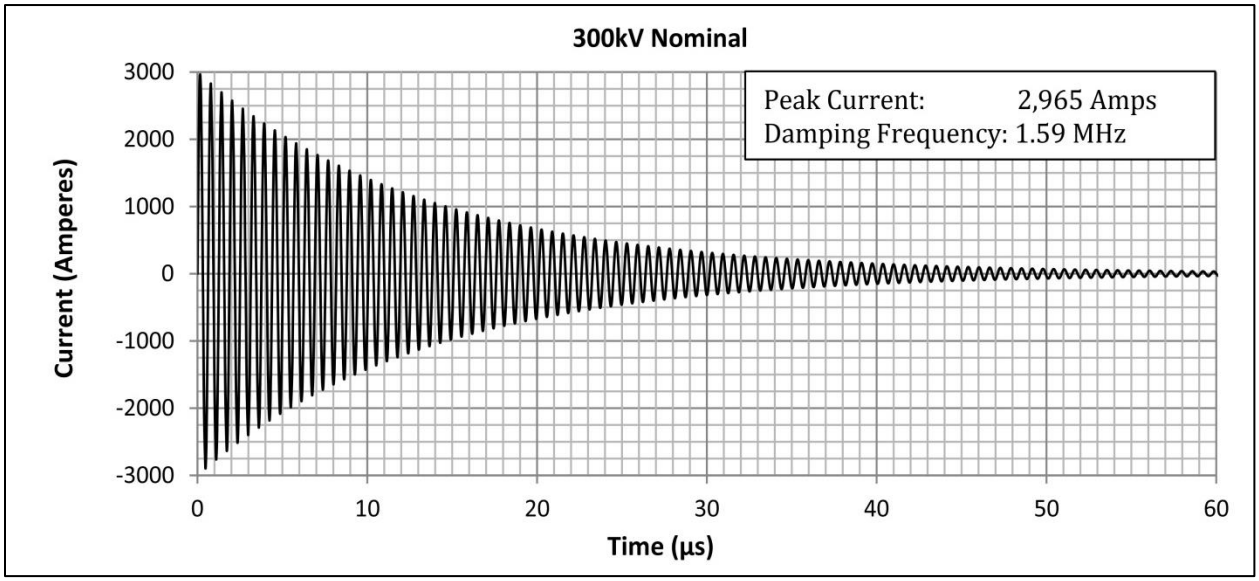


FIGURE 9. HESD nominal limit current waveform

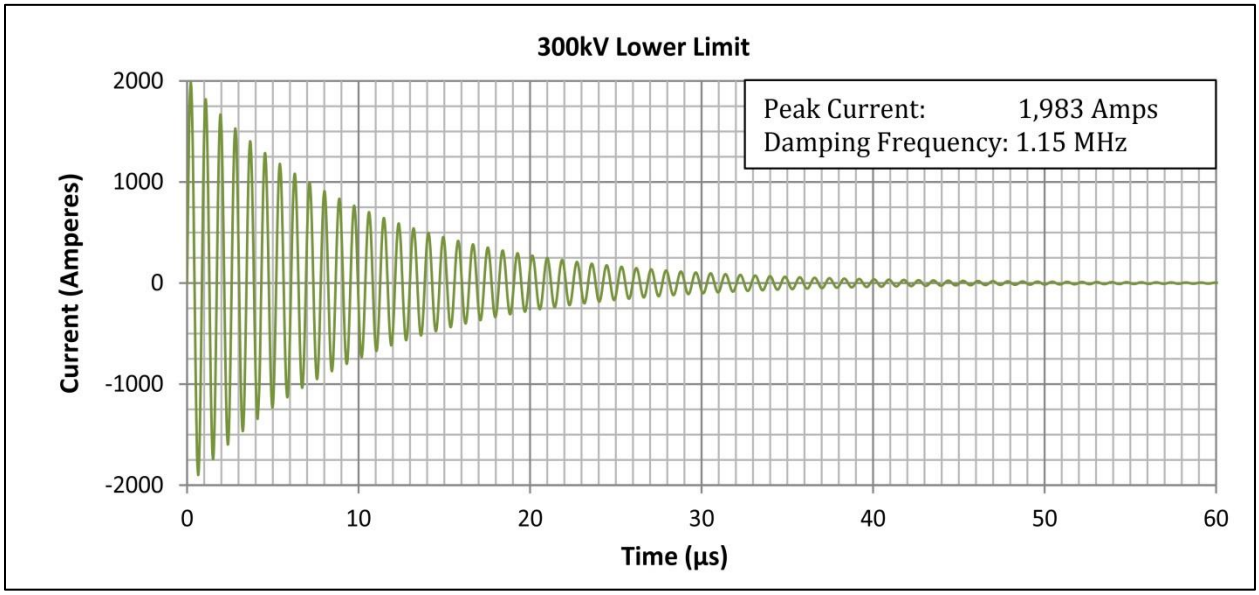


FIGURE 10. HESD lower limit current waveform

The calibration test should be a shorted discharge to the ground plane from the test electrode using a current transformer to measure the current for the man-carry/hot tube loaded configuration, and a shorted discharge to the toroid from the test ground electrode for the VERTREP/sling load/external carry configuration.

At a minimum, the output from the test equipment should be verified and recorded on a daily basis prior to testing and at the conclusion of testing for the day in accordance with the calibration limits of FIGURE 8 through FIGURE 10. If a salient is used on the test item, it should be considered part of the discharge circuit and used during the daily verification.

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All measurement equipment must be calibrated for the test data to be valid. Equipment claiming MIL-STD-331 or MIL-STD-464 compliance does not guarantee the equipment actually meets the performance requirements of those documents and full compliance should be verified.

Any attenuation used in the measurement circuit must be calibrated and clearly identified on the data plots to verify the results.

5.8 Test Sample Quantities.

5.8.1 AUR. The minimum sample size for AUR HESD testing should be 10 items for all categories of ordnance. No inert testing is allowed unless the device/AUR or the subcomponent of the AUR has been verified as a Faraday cage (per 4.5.1 and 5.5.1) or has previously been tested live in accordance with 5.3 of this document. Fewer than ten sets of EIDs are allowed for testing, but the equivalent number of discharges delivered must come to 10 samples. This increases the possibility of failure due to additional discharges to the reduced sample set.

5.8.2 Bare/Man-Carry Devices. No inert testing is allowed unless the item has been verified as a Faraday cage in accordance with the requirements of this document or has previously been tested live in accordance with 5.3 of this document.

5.8.3 Incremental Voltage Testing. A minimum of four devices of the 10 samples required for HESD testing in the tested configuration (bare/man-carry, hot tube loaded, external carry, or VERTREP/sling load) must receive incremental voltage testing of 50 kV, 100 kV, 150 kV, 200 kV, 250 kV, and 300 kV. Even for the EIDs that receive incremental testing, all EIDs must receive both the + and – 300 kV discharge. To document the safety and operational performance requirements of the item to all possible HESD exposures, conduct testing using the component requirements of TABLE VII and the appropriate electrical test equipment schematic from FIGURE 6 for bare/man-carry/hot tube loaded configuration or FIGURE 7 for VERTREP/ sling load/external carry configuration. One of the devices will be tested using positive voltages starting at 50 kV and increasing the voltage up to 300 kV in 50 kV steps (and must also include a – 300 kV discharge), and one other device should be tested starting at 300 kV and decreasing the voltage in 50 kV steps down to 50 kV (and must also include a – 300 kV discharge). The other two devices should be tested using negative voltages in a similar incremental voltage fashion (one device starting at – 50 kV and decreasing the voltage to – 300 kV in 50 kV steps (and must also include a +300 kV discharge), and the other device starting at – 300 kV and increasing the voltage in 50 kV steps up to – 50 kV (and must also include a +300 kV discharge)). Irrespective of the sample size used for the incremental voltage testing, the equivalent number of discharges delivered to four samples must be delivered to the sample size used for this testing (see TABLE VIII as an example). The four devices required for this testing should be from the 10 total samples required for testing.

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TABLE VIII. Example of Incremental Voltage HESD Testing On A Reduced Sample Set (4 sets of EIDs)

HESD Test Sequences 1-6		
Sequence No.[1]	**Test Set[2,3]**	**ESD Environment**
1	Set 1	+50 kV
1	Set 1	+100 kV
1	Set 1	+150 kV
1	Set 1	+200 kV
1	Set 1	+250 kV
1	Set 1	+300 kV
1	Set 1	-300 kV
2	Set 1	-300 kV
2	Set 1	+300 kV
3	Set 1	-300 kV
3	Set 1	+300 kV
4	Set 2	+300 kV
4	Set 2	+250 kV
4	Set 2	+200 kV
4	Set 2	+150 kV
4	Set 2	+100 kV
4	Set 2	+50 kV
4	Set 2	-300 kV
5	Set 2	-300 kV
5	Set 2	+300 kV
6	Set 2	-300 kV
6	Set 2	+300 kV

HESD Test Sequences 7-10		
Sequence No.[1]	**Test Set[2,3]**	**ESD Environment**
7	Set 3	-50 kV
7	Set 3	-100 kV
7	Set 3	-150 kV
7	Set 3	-200 kV
7	Set 3	-250 kV
7	Set 3	-300 kV
7	Set 3	+300 kV
8	Set 3	+300 kV
8	Set 3	-300 kV
9	Set 4	-300 kV
9	Set 4	-250 kV
9	Set 4	-200 kV
9	Set 4	-150 kV
9	Set 4	-100 kV
9	Set 4	-50 kV
9	Set 4	+300 kV
10	Set 4	+300 kV
10	Set 4	-300 kV

NOTES:
1. The Test Sequence includes all discharge points.
2. All electronic circuits/firing circuits and components must be verified to function normally prior to and after HESD testing of each set of EIDs.
3. Prior to and after the HESD test, bridgewire resistance measurements must be measured for each set of EIDs.

5.9 Test Item Configuration. If the device or AUR contains electrically initiated devices and/or electrically initiated devices connected to an external connector of the AUR that have not been tested to the requirements of this document, then testing with live EIDs is required. Bare device/AUR with live EIDs in the man-carry or hot tube loaded configuration will be tested using direct contact discharge only; a packaged device or AUR with live EIDs in the VERTREP/sling load or external carry configuration will be tested using the approaching probe discharge method only unless the EIDs are internal to the AUR and meet the Faraday cage protection requirements as defined in this document.

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5.9.1 EIDs. EIDs contained in a non-Faraday cage AUR and not connected to an exterior electrical connector, or with different circuitry leading to the EID than a previously tested version, and which have not previously been tested in the same AUR configuration must be tested with live devices in the AUR configuration.

Note: Operability testing of the AUR electronics, which evolved from a previously tested EID internal to the AUR may be performed using an inert/instrumented EID.

5.9.2 Percussion Initiated Devices. Percussion initiated devices not in a documented Faraday cage or other non-electric devices must be tested live. Test configuration should be bare for man-carry/hot tube loaded evaluation, in the shipping configuration for VERTREP/sling load evaluation, and in the installed configuration with cabling attached for the external carry evaluation.

5.9.3 Configuration Changes. Any change to a previously HESD tested and/or certified device or AUR, including changes to component materials, energetic materials, EIDs, packaging/shipping container, electrical interfaces, communication interfaces, or remote devices should be subjected to HESD testing to verify the new configuration meets the requirements of this document and assigned a new nomenclature/cataloging number.

5.9.4 Instrumented Testing. Instrumented testing should not be performed on electrically initiated devices that have not been tested live either at the device level or in the AUR configuration as defined in 5.3 until a Joint Service agreement on instrumentation methods and procedures are developed. Additional investigation is needed to develop a Joint Service instrumented testing methodology that will provide the safety and operability data for an EID that will include the specific instrumentation, data acquisition, and safety and operability margins for HESD testing. Instrumented testing is allowed for items where the electrical/firing interface has changed, but no changes were made to the EID.

5.10 HESD Test Procedure.

5.10.1 Bare/Man-Carry Discharges. For man-carry and hot tube loading exposure, direct contact discharges to all electrical interfaces should be made to each pin on an external connector that connects directly or indirectly to an EID, firing circuit, control interface, or any other ordnance related function such as antennas or inductive antennas. All plastic and non-metallic components of a device or AUR that contain ordnance, circuitry that controls the ordnance, or provides a path to the internal components of an AUR should receive direct contact discharges. Any other possible entry points including joints between sections and covers over cabling should also be tested. Salients must be used to direct the discharge to test points where arcing could occur to adjacent surfaces such as electrical pins on a connector. If the item is bare energetic material, the material must be in a container of similar size as the test sample and it should be isolated from the ground plane to preclude arcing around the energetic material. The energetic sample host container or test fixture must accommodate the salients that deliver the ESD pulse to the sample. The host container or test fixture should be configured to replicate or exceed the maximum ESD pulse power transfer to the energetic when in its ordnance configuration.

5.10.2 VERTREP/Sling Load Discharges. For VERTREP/sling load exposure, approaching probe discharges should be to any point on the non-Faraday cage shipping container

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near any electrical interface, firing circuit, EID, antenna, plastic component, or ordnance material that may provide a path for the discharge energy to the ordnance item or AUR. Salients must be used to direct the discharge to test points where arcing could occur to adjacent surfaces. EIDs must be subjected to no-fire and all fire testing in accordance with the requirements in MIL-DTL-23659F after the VERTREP/sling load test to verify conformance with safety and operability specifications of the device.

5.10.3 External Carry Discharges. For external carry exposure, items should be in/on the launcher and should be tested with all electrical cables attached; approaching probe discharges should be to any point on the installed external carry configuration item near any electrical interface, firing circuit, EID, antenna, plastic component, metal joint, or ordnance material that may provide a path for the discharge energy to the ordnance item or AUR. Salients must be used to direct the discharge to test points where arcing could occur to adjacent surfaces. EIDs must be subjected to no-fire and all-fire testing in accordance with the requirements in MIL-DTL-23659F after the external carry test to verify conformance with safety and operability specifications of the device.

5.10.4 Hot Tube Loaded Discharges. For hot tube loaded exposure, items should be tested with direct contact HESD discharges to all external electrical pins, antennas, and non-metallic points of entry that may provide a path for the discharge energy to the ordnance item or AUR. Salients must be used to direct the discharge to test points where arcing could occur to adjacent surfaces. EIDs must be subjected to no-fire and all fire testing in accordance with the requirements in MIL-DTL-23659F after the hot tube loaded test to verify conformance with safety and operability specifications of the device.

5.10.5 Remote Control Item Testing. Items with a remote control should be tested as individual components (transmitter and receiver), and as a system with both components operating to determine if a discharge to one of the system components can cause a safety or operability consequence in the other component.

5.10.6 Test Sequences. An equivalent of 10 test sequences should be conducted when testing to the helicopter-borne ESD threat. A test sequence is defined as a series of discharges to the device or AUR in accordance with the requirements of TABLE VII (2 minimum discharges per test point) at the test points identified in the pre-test assessment (5.4 of this document). Subsequent sequences may be conducted by using a different device/AUR or on the same device/AUR with a different set of EIDs and electronic/electrical subsystems. If the subsequent testing uses the same device or AUR with the same EIDs, functional verification and bridgewire resistance testing should be performed after each series of discharges as defined above, unless documented in the test plan.

5.10.7 Arming. Operational conditions may require that a bare/man-carry armed item be handled or carried when in or around operating rotary winged aircraft; therefore, HESD testing should include testing in the armed condition for this configuration without regard to operational procedures.

5.10.8 Bridgewire Resistance Measurements. Prior to and after the HESD test, bridgewire resistance measurements must be made for all live EIDs in the device or AUR.

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5.10.9 No-Fire and All-Fire Verification. All EIDs subjected to live HESD testing should be evaluated to ensure they still meet the specified device No-Fire and All-Fire parameters. This verification is not a re-establishment of the No-Fire and All-Fire values. The verification testing is intended to evaluate if the device will not fire when subjected to the specified No-Fire stimulus and will produce its expected output when subjected to the specified All-Fire stimulus. This validates that the device has not dudded and has not had a change in sensitivity as a result of HESD testing.

5.10.10 Verification of Circuit Functionality. All electronic circuits/firing circuits and components should be verified to function normally prior to and after HESD testing.

5.10.11 Temperature and Humidity. The test should be conducted on devices at an ambient temperature of +23°C±10°C (+73°F±18°F). Relative humidity of the test chamber should be no greater than 50 percent. The device should be preconditioned at +23°C±3°C (+73°F± 5°F), relative humidity no greater than 50 percent for no less than 24 hours prior to testing. If the size of the AUR precludes indoor controlled testing, test should be performed at the lowest humidity possible, but as a minimum, there should be no moisture on the test item (non-condensing humidity level). It should be noted that ESD effects on ordnance at a low temperature/humidity may be different from the standard test temperature/humidity. APPENDIX C explains this phenomenon.

5.11 Post HESD Test Requirements.

5.11.1 No-Fire Safety Requirement. All EIDs subjected to HESD testing should be subjected to bridgewire resistance measurement and no-fire performance testing in accordance with MIL-DTL-23659F after HESD testing to verify the item still meets its no-fire (safety) requirement. Following the no-fire testing, these items should be subjected to all-fire testing in accordance with the all-fire specification to verify the item has not dudded.

5.11.2 Inspection. All EIDs and percussion-initiated devices not in a documented Faraday cage or other non-electric devices should be inspected for initiation and should be functionally tested after HESD exposure to verify proper operation.

5.11.3 Operability. A complete checkout/functional test of the device or AUR to include all electrical interfaces should be completed after each set of discharges to verify the item is functional and was not damaged or affected by the HESD test discharges.

Items with a remote control should be verified to function normally after each set of discharges. Both transmit and receive functions should be tested and verified as operational.

Any functional or operational anomaly that prevents the device or AUR from completing its function when commanded after the selected number of HESD discharges is considered an operability failure.

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5.11.4 Data Collection. Record the following for each test:

a. Stimulus waveform parameter data/plots.

b. Test configuration/test mode.

c. Test discharge point.

d. Post-discharge functional check pass/fail results.

e. Visual damage inspection results.

f. Test item responses (e.g., components affected, visible reactions, burning, upsets, recovery times, etc.)

g. EID bridgewire resistance and determination of degradation, if required.

h. Date, time, location.

5.12 HESD Test Assessment and Analysis Reports.

5.12.1 Data Sharing. In order to facilitate data sharing across the Services and to ensure test data consistency and uniform operational guidance, an ESD test report should be developed as part of every ESD test. All ESD test reports should be available for inclusion in a centralized DoD database.

5.12.2 Test Report. Test reports should contain the following information and documentation as a minimum:

5.12.2.1 Drawing Package. A complete drawing package for the device or AUR identifying all materials of construction and all electrical interfaces.

5.12.2.2 HESD Test Record. The test record consists of the following:

a. Date(s) of test.

b. Test location/facility.

c. Test personnel names.

d. Test and conditioning records documenting compliance with the following:

1) All test equipment/instrumentation used to measure the waveforms, type/model/serial number, equipment/instrumentation calibration data, and including the calibration waveforms as specified in 5.7.

2) Configuration of test item, Mark (MK)/MOD, nomenclature/DODIC, model, drawing, part number, revision, NSN, as applicable. The description should include any variations of the test item from the final configuration of the live

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ordnance item. The configuration should also include a complete description of the test item for HESD testing including if it is powered on, armed, electrical interface pins designated for each test, etc.

3) Test item record: Each test article should be identified and described by a test item record. The record should include pre-test performance, performance during the test, and post-test performance information and documents as described below.

5.12.2.3 Pre-test Performance. The operation/function of the device or AUR should be verified and documented in accordance with 5.10 prior to HESD testing to provide a reference for the operability assessment. This includes any functional tests required to verify proper operation of the device or AUR, including operation of remote-controlled devices. A record should be made of all data to determine compliance with required performance and, when applicable, to provide a reference level or criteria for checking desired performance of the test item during or at the conclusion of the test. If several tests are to be performed in sequence and the cumulative effect of use conditions is desired, then the measurement of performance level prior to each individual test may be deleted and only the pre-sequence measurement performed. The pre-test performance check may be made after installation of the item under test if installation conditions necessitate it. Pre-HESD test bridgewire resistance measurements are part of the pre-test performance data as specified in 5.10.8.

5.12.2.4 Performance During Test. When operation of the test item is required during the test, a record should be kept of the data for comparison with pre-test or post-test performance as required. The conditions during the performance check should be those specified in the individual test.

5.12.2.5 Post-test Performance. The operation of the test item should be verified and documented at the conclusion of the test as specified in 5.3.6 and 5.11.3, and compared with pre-test performance or during-test performance, whichever is required for determining the operability of the item after HESD testing. This functional test should include the operation and status of any remote-controlled device(s). Also, post-HESD test bridgewire resistance measurements, no-fire test results, and all-fire test results are part of the post-test performance documentation that must be included.

5.12.2.6 Photographs. Photographs of each test configuration and test point with any salients used to direct HESD discharges to the specific test points.

5.12.2.7 Test Results. All test results, data, and ESD calibration waveforms including reduced voltage test results.

5.12.2.8 Analysis of Test Results and Data. Test data should be compared and evaluated against the criteria set forth in MIL-STD-464, MIL-STD-331, the test item specification, and test item test plan to determine if the item met the required criteria. Deficiencies should be identified and documented.

5.12.3 Conclusion on the ESD Sensitivity of the Item and Criteria for Assigning the HESD Code. FIGURE 5 provides a sample of the ESD Summary datasheet.

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5.12.3.1 Device Condition After HESD Tests. At the completion of these tests, the ordnance should be safe for transportation, storage, handling and use, as well as operable during and after any possible S4 HESD exposure. Safety and operability failures are evaluated separately. No safety failures are allowed for bare/man-carry, hot tube loaded items, external carry or packaged items.

5.12.3.2 Safe for Use. In accordance with 5.2.4, in order for the ordnance item to be classified as Safe for Use, it must not have been dudded or initiated during the HESD test. Ordnance use includes installation, firing or release of the ordnance. This requirement also includes passing the reduced voltage tests.

5.12.3.3 Operable. In accordance with 5.2.4, when the ordnance is provided its required inputs, it should perform to completion its function and sequence producing all required outputs within the operating period or at the specified time after any possible HESD exposure. This requirement includes passing the reduced voltage tests also.

5.12.3.4 Not Safe and Operable. If the test item is not both safe and operable as described above for any possible HESD exposure (man-carry, hot tube loaded, VERTREP/sling load, and external carry), it is sensitive to one or more HESD exposures.

5.12.3.5 Assignment of Joint Service HESD Code. At the conclusion of all testing and after analysis of all results, the ordnance should be assigned a HESD Code as delineated in TABLE IX:

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TABLE IX. HESD Code Descriptions

Code	Description
HS	Safe and operable for HESD exposure. This indicates that the ordnance is both safe and operable for HESD exposures including man-carry/hot tube loading, VERTREP/sling load, and external carry. It has passed live HESD testing to all EIDs and/or energetic material with no dudding or initiation, and has passed HESD testing to all electrical interfaces that control or interface to the item with no effect on performance or functioning as described in 5.12.3.2 and 5.12.3.3 above.
HV	Safe and operable for VERTREP/sling load HESD exposure. This indicates that the ordnance is both safe and operable after VERTREP/sling load HESD exposure only. It has passed live HESD testing in the shipping configuration with no dudding or initiation with no effect on performance or functioning as described in 5.12.3.2 and 5.12.3.3 above. All items must be tested or analyzed to at least the VERTREP/sling load requirements.
HE	Safe and operable for external carry HESD exposure. This indicates that the ordnance is both safe and operable for external carry and VERTREP/sling load HESD exposures only. It has passed live HESD testing in the external carry configuration with no dudding or initiation, and with no effect on performance or functioning as described in 5.12.3.2 and 5.12.3.3 above. The ordnance has also been evaluated or tested for VERTREP/sling load exposure in the shipping configuration also.
HX	Energetic susceptibility or operable susceptibility for HESD exposure. This indicates that the ordnance failed live HESD testing or was not tested and evaluated to one or more EIDs or energetic material with either dudding or initiation, or both. In addition, it failed or was not tested and evaluated for one or more operable requirements and may not perform or function as designed. This code must only be assigned for items not for service use in the field, or when the program has a declaration statement that HESD is not part of its operating environment.

5.12.4 Safety Data Package. The results of all HESD testing and analysis conducted relevant to assessing the ordnance item/munitions safety for service should be compiled into a safety data package for review. The package must include the previously-approved test plan, including the rationale for any variance from the joint requirements. In addition, any deviation from that approved plan should be presented along with an analysis showing why the results should be accepted. The package must also provide any safety and operability deductions derived from those results.

6. DOCUMENTATION

Test documentation incorporates the total assimilation of a test program. Without proper documentation, a test program could be forfeited. The test plan lays the groundwork for accomplishing the mission of certification/compliance. All the minute details for conducting the test should be captured here. The substance of the test program resides in adherence to the test procedures during the test and in the data collection process. Faithful execution of the test procedures and explicit data collection will bring the test goals into focus and lead to a successful completion of the test mission. The test report unifies the goals and objectives by summing up

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the test data into viable conclusions and recommendations, thus finalizing the process. The documentation for test reporting is detailed in 4.15 for PESD and 5.12 for HESD.

7. NOTES

7.1 Intended Use. This handbook can be used by any facility engaged in ESD testing for any of the military departments.

7.2 Supersession Data. Not applicable.

7.3 Cross-Reference of Classification and Substitutability Data. Not applicable.

7.4 Subject Term (Keyword) Listing.

Electromagnetic (EM)
Electromagnetic Compatibility (EMC)
Electromagnetic Environmental Effects (E3)
Electromagnetic Radiation Hazards
Electrostatic Discharge (ESD)
Helicopter-borne Electrostatic Discharge (HESD)
Personnel-borne Electrostatic Discharge (PESD)

7.5 Identification of Changes. Not applicable.

7.6 Technical Points of Contact. Requests for additional information or assistance on this standard can be obtained from the following:

Army

US Army Test and Evaluation Command (HQ, ATEC)
6617 Aberdeen Blvd., Building 2202
ATTN: CSTE-OPS
Aberdeen Proving Ground, MD 21005-5001
(443) 861-9664
usarmy.apg.atec.mbx.atec-hq-g3-5@mail.mil

Army Test and Evaluation Command (ATEC)
E3 Test Division (E3 for Missile System Functions)
ATTN: TERT-ECE
Building 8978
Redstone Arsenal, AL 35898-8052
(256) 313-6970
usarmy.redstone.atec.list.rtc-jocg@mail.mil

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Army Test and Evaluation Command (ATEC)
White Sands Test Center
Survivability, Vulnerability and Assessment Directorate
ATTN: TEWS-SVE
21225 Headquarters Avenue
WSMR, NM 88002
(575) 678-6307
usarmy.wsmr.atec.list.datts-sv-e@mail.mil

Marine Corps

Command Safety (00T)
Marine Corps Systems Command (MARCORSYSCOM)
ATTN: 00T
2200 Lester Street
Quantico, VA 22134-6050
(703) 432-4930
mcsc_safety@usmc.mil

Navy

Commanding Officer
Naval Ordnance Safety and Security Activity
ATTN: N8
3817 Strauss Ave, Suite 108
Indian Head, MD 20640-5151
(301) 744-6003
nossa_esd@us.navy.mil

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APPENDIX A
CALCULATIONS FOR PESD WITH NOMINAL VALUES OF 25KV, 500 OHMS, AND 500 PF

(All calculations shown are positive; negative waveforms are inverted)

Time span and initial condition values:

$t = (0, \quad .001\mu s, \quad 1.4\mu s) \quad i_{initial} = 0\ A$

Nominal values for voltage, resistance, inductance, and capacitance:

$R_{nom} = 500\ \Omega \quad C_{nom} = 500\ pF \quad L_{nom} = 3\ \mu H \quad V_{nom} = 25000\ V$

Calculations for alpha and omega based on nominal values:

$$\alpha_{nom} = \frac{R_{nom}}{2 \cdot L_{nom}} \qquad \omega_{nom} = \frac{1}{\sqrt{L_{nom} \cdot C_{nom}}}$$

$$\alpha_{nom} = (8.333 \cdot 10^7)\ \frac{1}{s} \qquad \omega_{nom} = (2.582 \cdot 10^7)\frac{1}{s}$$

Calculations for S1 and S2 based on nominal values of alpha and omega:

$$s_{1nom} = -\alpha_{nom} + \sqrt{{\alpha_{nom}}^2 - {\omega_{nom}}^2} \qquad s_{2nom} = -\alpha_{nom} - \sqrt{{\alpha_{nom}}^2 - {\omega_{nom}}^2}$$

$$s_{1nom} = -4.101 \cdot 10^6\ \frac{1}{s} \qquad s_{2nom} = -1.626 \cdot 10^8\ \frac{1}{s}$$

Calculations for A_{1nom} and A_{2nom} based on nominal values:

$$A_{2nom} = \frac{\left(\frac{-1}{L_{nom}} \cdot (R_{nom} \cdot i_{initial} + V_{nom})\right) - (i_{initial} \cdot s_{2nom})}{(s_{2nom} - s_{1nom})}$$

$$A_{2nom} = 52.588A$$

$$A_{1nom} = i_{initial} - A_{2nom}$$

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$A_{1nom} = -52.588A$

Double exponential equation for current & voltage waveforms based on nominal value calculations:

$i_{nom}(t) = -\left(A_{1nom} \cdot e^{(s_{1nom} \cdot t)} + A_{2nom} \cdot e^{(s_{2nom} \cdot t)}\right)$

$V_{nom}(t) = i_{nom}(t) \cdot R_{nom}$

Upper Limit values for voltage (maximum), resistance (minimum), inductance (minimum), and capacitance (maximum):

$R_{min} = R_{nom} - (R_{nom} \cdot 0.05)$ $C_{max} = C_{nom} + (C_{nom} \cdot 0.05)$ $L_{min} = 1\ \mu H$ $V_{max} = 25500\ V$

$R_{min} = 475\ \Omega$ $C_{max} = 525\ pF$

Calculations for alpha and omega based on Upper Limit values:

$$\alpha_{upperLimit} = \frac{R_{min}}{2 \cdot L_{min}} \qquad \omega_{upperLimit} = \frac{1}{\sqrt{L_{min} \cdot C_{max}}}$$

$$\alpha_{upperLimit} = (2.375 \cdot 10^8)\frac{1}{s} \qquad \omega_{upperLimit} = (4.364 \cdot 10^7)\frac{1}{s}$$

Calculations for S1 and S2 based on Upper Limit values of alpha and omega:

$$s_{1upperLimit} = -\alpha_{upperLimit} + \sqrt{\alpha_{upperLimit}^2 - \omega_{upperLimit}^2} \qquad s_{2upperLimit} = -\alpha_{upperLimit} - \sqrt{\alpha_{upperLimit}^2 - \omega_{upperLimit}^2}$$

$$s_{1upperLimit} = -4.044 \cdot 10^6\ \frac{1}{s} \qquad s_{2upperLimit} = -4.71 \cdot 10^8\ \frac{1}{s}$$

Calculations for A1 and A2 based on Upper Limit values listed above:

$$A_{2upperLimit} = \frac{\left(\frac{-1}{L_{min}} \cdot (R_{min} \cdot i_{initial} + V_{max})\right) - (i_{initial} \cdot s_{2upperLimit})}{(s_{2upperLimit} - s_{1upperLimit})}$$

$A_{2upperLimit} = 54.614\ A$

$A_{1upperLimit} = i_{initial} - A_{2upperLimit}$

$A_{1upperLimit} = -54.614\ A$

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Double exponential equation for current & voltage waveforms based on upper Limit value calculations:

$$i_{upperLimit}(t) = -\left(A_{1upperLimit} \cdot e^{(s_{1upperLimit} \cdot t)} + A_{2upperLimit} \cdot e^{(s_{2upperLimit} \cdot t)}\right)$$

$$V_{upperLimit}(t) = i_{upperLimit}(t) \cdot R_{min}$$

Lower Limit values for voltage (maximum), resistance (minimum), inductance (minimum), and capacitance (maximum):

$R_{max} = R_{nom} + (R_{nom} \cdot 0.05)$ $C_{min} = C_{nom} - (C_{nom} \cdot 0.05)$ $L_{max} = 5\mu H$ $V_{min} = 24500V$

$R_{max} = 525\Omega$ $C_{min} = 475\ pF$

Calculations for alpha and omega based on Lower Limit values:

$$\alpha_{lowerLimit} = \frac{R_{max}}{2 \cdot L_{max}} \qquad \omega_{lowerLimit} = \frac{1}{\sqrt{L_{max} \cdot C_{min}}}$$

$$\alpha_{lowerLimit} = (5.25 \cdot 10^7)\frac{1}{s} \qquad \omega_{lowerLimit} = (2.052 \cdot 10^7)\frac{1}{s}$$

Calculations for S1 and S2 based on Lower Limit values of alpha and omega:

$$s_{1lowerLimit} = -\alpha_{lowerLimit} + \sqrt{\alpha_{lowerLimit}^2 - \omega_{lowerLimit}^2} \qquad s_{2lowerLimit} = -\alpha_{lowerLimit} - \sqrt{\alpha_{lowerLimit}^2 - \omega_{lowerLimit}^2}$$

$$s_{1lowerLimit} = -4.176 \cdot 10^6\ \frac{1}{s} \qquad s_{2lowerLimit} = -1.008 \cdot 10^8\ \frac{1}{s}$$

Calculations for A1 and A2 based on Lower Limit values:

$$A_{2lowerLimit} = \frac{\left(\frac{-1}{L_{max}} \cdot (R_{max} \cdot i_{initial} + V_{min})\right) - (i_{initial} \cdot s_{2lowerLimit})}{(s_{2lowerLimit} - s_{1lowerLimit})}$$

$$A_{2lowerLimit} = 50.7A$$

$$A_{1lowerLimit} = i_{initial} - A_{2lowerLimit}$$

$$A_{1lowerLimit} = -50.7A$$

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Double exponential equation for current & voltage waveforms based on Lower Limit value calculations:

$$i_{lowerLimit}(t) = -\left(A_{1lowerLimit} \cdot e^{(s_{1lowerLimit} \cdot t)} + A_{2lowerLimit} \cdot e^{(s_{2lowerLimit} \cdot t)}\right)$$

$$V_{lowerLimit}(t) = i_{lowerLimit}(t) \cdot R_{max}$$

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APPENDIX B

CALCULATIONS FOR PESD WITH NOMINAL VALUES OF 25KV, 5000 OHMS, AND 500 PF

(All calculations shown are positive; negative waveforms are inverted)

Time span and initial condition values:

$t = (0, \quad .0003\mu s, \quad .13\mu s) \qquad i_{initial} = 0A$

Nominal values for voltage, resistance, inductance, and capacitance:

$R_{nom} = 5000\ \Omega \quad C_{nom} = 500\ pF \quad L_{nom} = 3\ \mu H \quad V_{nom} = 25000\ V$

Calculations for alpha and omega based on nominal values:

$$\alpha_{nom} = \frac{R_{nom}}{2 \cdot L_{nom}} \qquad \omega_{nom} = \frac{1}{\sqrt{L_{nom} \cdot C_{nom}}}$$

$$\alpha_{nom} = (8.333 \cdot 10^8)\ \frac{1}{s} \qquad \omega_{nom} = (2.582 \cdot 10^7)\frac{1}{s}$$

Calculations for S1 and S2 based on nominal values of alpha and omega:

$$s_{1nom} = -\alpha_{nom} + \sqrt{{\alpha_{nom}}^2 - {\omega_{nom}}^2} \qquad s_{2nom} = -\alpha_{nom} - \sqrt{{\alpha_{nom}}^2 - {\omega_{nom}}^2}$$

$$s_{1nom} = -4.001 \cdot 10^5\ \frac{1}{s} \qquad s_{2nom} = -1.666 \cdot 10^9\ \frac{1}{s}$$

Calculations for A1 and A2 based on nominal values:

$$A_{2nom} = \frac{\left(\frac{-1}{L_{nom}} \cdot (R_{nom} \cdot i_{initial} + V_{nom})\right) - (i_{initial} \cdot s_{2nom})}{(s_{2nom} - s_{1nom})}$$

$$A_{2nom} = 5.002\ A$$

$$A_{1nom} = i_{initial} - A_{2nom}$$

$$A_{1nom} = -5.002\ A$$

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Double exponential equation for current & voltage waveforms based on nominal value calculations:

$$i_{nom}(t) = -\left(A_{1nom} \cdot e^{(s_{1nom} \cdot t)} + A_{2nom} \cdot e^{(s_{2nom} \cdot t)}\right)$$

$$V_{nom}(t) = i_{nom}(t) \cdot R_{nom}$$

Upper Limit values for voltage (maximum), resistance (minimum), inductance (minimum), and capacitance (maximum):

$R_{min} = R_{nom} - (R_{nom} \cdot 0.05)$ $C_{max} = C_{nom} + (C_{nom} \cdot 0.05)$ $L_{min} = 1\ \mu H$ $V_{max} = 25500\ V$

$R_{min} = 4750\ \Omega$ $C_{max} = 525\ pF$

Calculations for alpha and omega based on Upper Limit values:

$$\alpha_{upperLimit} = \frac{R_{min}}{2 \cdot L_{min}} \qquad \omega_{upperLimit} = \frac{1}{\sqrt{L_{min} \cdot C_{max}}}$$

$$\alpha_{upperLimit} = (2.375 \cdot 10^9)\frac{1}{s} \qquad \omega_{upperLimit} = (4.364 \cdot 10^7)\frac{1}{s}$$

Calculations for S1 and S2 based on Upper Limit values of alpha and omega:

$$s_{1upperLimit} = -\alpha_{upperLimit} + \sqrt{\alpha_{upperLimit}^2 - \omega_{upperLimit}^2} \qquad s_{2upperLimit} = -\alpha_{upperLimit} - \sqrt{\alpha_{upperLimit}^2 - \omega_{upperLimit}^2}$$

$$s_{1upperLimit} = -4.01 \cdot 10^5\ \frac{1}{s} \qquad s_{2upperLimit} = -4.75 \cdot 10^9\ \frac{1}{s}$$

Calculations for A1 and A2 based on Upper Limit values listed above:

$$A_{2upperLimit} = \frac{\left(\frac{-1}{L_{min}} \cdot (R_{min} \cdot i_{initial} + V_{max})\right) - (i_{initial} \cdot s_{2upperLimit})}{(s_{2upperLimit} - s_{1upperLimit})}$$

$A_{2upperLimit} = 5.369A$

$A_{1upperLimit} = i_{initial} - A_{2upperLimit}$

$A_{1upperLimit} = -5.369\ A$

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Double exponential equation for current & voltage waveforms based on upper Limit value calculations:

$$i_{upperLimit}(t) = -\left(A_{1upperLimit} \cdot e^{(s_{1upperLimit} \cdot t)} + A_{2upperLimit} \cdot e^{(s_{2upperLimit} \cdot t)}\right)$$

$$V_{upperLimit}(t) = i_{upperLimit}(t) \cdot R_{min}$$

Lower Limit values for voltage (minimum), resistance (maximum), inductance (maximum), and capacitance (minimum):

$R_{max} = R_{nom} + (R_{nom} \cdot 0.05)$ $C_{min} = C_{nom} - (C_{nom} \cdot 0.05)$ $L_{max} = 5\mu H$ $V_{min} = 24500V$

$R_{max} = 5250\Omega$ $C_{min} = 475\ pF$

Calculations for alpha and omega based on Lower Limit values:

$$\alpha_{lowerLimit} = \frac{R_{max}}{2 \cdot L_{max}} \qquad \omega_{lowerLimit} = \frac{1}{\sqrt{L_{max} \cdot C_{min}}}$$

$$\alpha_{lowerLimit} = (5.25 \cdot 10^8)\frac{1}{s} \qquad \omega_{lowerLimit} = (2.052 \cdot 10^7)\frac{1}{s}$$

Calculations for S1 and S2 based on Lower Limit values of alpha and omega:

$$s_{1lowerLimit} = -\alpha_{lowerLimit} + \sqrt{\alpha_{lowerLimit}^2 - \omega_{lowerLimit}^2} \qquad s_{2lowerLimit} = -\alpha_{lowerLimit} - \sqrt{\alpha_{lowerLimit}^2 - \omega_{lowerLimit}^2}$$

$$s_{1lowerLimit} = -4.012 \cdot 10^5\ \frac{1}{s} \qquad s_{2lowerLimit} = -1.05 \cdot 10^9\ \frac{1}{s}$$

Calculations for A1 and A2 based on Lower Limit values:

$$A_{2lowerLimit} = \frac{\left(\frac{-1}{L_{max}} \cdot (R_{max} \cdot i_{initial} + V_{min})\right) - (i_{initial} \cdot s_{2lowerLimit})}{(s_{2lowerLimit} - s_{1lowerLimit})}$$

$$A_{2lowerLimit} = 4.67A$$

$$A_{1lowerLimit} = i_{initial} - A_{2lowerLimit}$$

$$A_{1lowerLimit} = -4.67\ A$$

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Double exponential equation for current & voltage waveforms based on Lower Limit value calculations:

$$i_{lowerLimit}(t) = -\left(A_{1lowerLimit} \cdot e^{(s_{1lowerLimit} \cdot t)} + A_{2lowerLimit} \cdot e^{(s_{2lowerLimit} \cdot t)}\right)$$

$$V_{lowerLimit}(t) = i_{lowerLimit}(t) \cdot R_{max}$$

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APPENDIX C
LOW TEMPERATURE/HUMIDITY EXPLORATORY TESTING OF ENERGETICS

The main section of MIL-HDBK-240 recommends consistent temperature and humidity (environmental) conditioning of ordnance at approximately standard conditions. The rationale is that the electrical properties of insulators/energetics may be affected by temperature/humidity. Thus, it is important to obtain repeatable results at/across various laboratories. Extreme temperature/humidity conditions are also costly to establish/maintain during conditioning. This appendix is concerned with the hazards of electrostatic discharge (ESD) in low temperature/ humidity environments. The contents of this section were borrowed from reference [1] presenting historical objective quality evidence in justifying the continuation of the existing ESD environment as per standards such as the JOTP-062. A brief description of the incident and ESD causality can be found on the NASA website [2].

When bare energetics (for example, propellants or explosives) are exposed to the PESD or HESD in cold/ dry environments it is advisable that their ESD susceptibility be evaluated for these environments.

Ordnance ESD environment was established starting in June 1985 largely due to the efforts of the ESD Hazards panel and propulsion systems hazards subcommittee formed jointly by several organizations, including Air Force Rocket Propulsion Laboratory (RPL), Johns Hopkins Applied Physics Laboratory (APL), U.S. Army Missile Command, Naval Weapons Center, TRW, Lockheed, Hercules Inc., Electromagnetic Applications Inc., and Los Alamos National Laboratories. The activities of ESD panel from 1985 to 1990 are explored in the document: “A Review of the ESD Panel Activities from June 1985 to March 1990,” reference [3]. The Pershing II incident on 11 January 1985 caused the death of three soldiers and the injury of sixteen others at Camp Redleg (Heilbronn), Germany [4]. The cause of the incident was determined to be ESD. The environment at the time of the incident was cold/dry. Between 1964 and 1987 there were eight (8) ESD related incidents documented by Johns Hopkins University APL. It appears that investigations that followed these incidents significantly impacted the accepted ordnance ESD environment.

Energetics sensitivity can vary significantly with temperature and humidity (references [5], [6], [7], and [8]). Propellants in the Pershing Incident were tested at ambient temperature and showed no reaction. However, ESD tests at low temperature (for example, conditions as per reference [9] -12˚C, 10˚F) did result in propellant reaction.

Sensitivity of propellants to cold temperatures and low humidity levels has been known as mentioned in reference [10]. Energetics that do not react at 70 degrees, for example, may react at significantly lower temperatures. The JOTP-062 does not currently address colder temperatures and low humidity conditions.

Additional information and guidance may be found on: https://e3.nswc.navy.mil an official Department of Navy website (access restricted, registration approval required).

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LOW TEMPERATURE/HUMIDITY EXPLORATORY TESTING OF ENERGETICS
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APPENDIX D
CALCULATIONS FOR HELICOPTER BORNE ESD (HESD) WITH NOMINAL VALUES OF 300KV, 1.5 OHMS, 10 UH, AND 1000 PF

(All calculations shown are positive; negative waveforms are inverted)

Time span and initial condition values:

$t = (0, \quad .01\mu s, \quad .60\mu s)$

$i_{initial} = 0A$

Nominal values for voltage, resistance, inductance, and capacitance:

$R_{nom} = 1.5\ \Omega \qquad C_{nom} = 1\ nF \qquad L_{nom} = 10\ \mu H \qquad V_{nom} = 300000V$

Calculations for alpha and omega based on nominal values:

$$\alpha_{nom} = \frac{R_{nom}}{2 \cdot L_{nom}} \qquad \omega_{onom} = \frac{1}{\sqrt{L_{nom} \cdot C_{nom}}}$$

$$\alpha_{nom} = (7.5 \cdot 10^4)\frac{1}{s} \qquad \omega_{onom} = (1 \cdot 10^7)\frac{1}{s}$$

Calculations for damping frequency based on nominal values of alpha and omega:

$$\omega_{dnom} = \sqrt{\frac{1}{L_{nom} \cdot C_{nom}} - \left(\frac{R_{nom}}{2 \cdot L_{nom}}\right)^2}$$

$$\omega_{dnom} = (10 \cdot 10^6)\frac{1}{s}$$

Calculations for A1 and A2 based on nominal values:

$A_{1nom} = V_{nom} \qquad A_{1nom} = (3 \cdot 10^5)V$

$A_{2nom} = \frac{\alpha_{nom} \cdot A_{1nom}}{\omega_{dnom}} \qquad A_{2nom} = (2.25 \cdot 10^3)V$

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Double exponential equation for current & voltage waveforms based on nominal values calculations:

$$i_{nom}(t)=\left((-c_{nom})\cdot(e^{-\alpha_{nom}\cdot t})\right)\cdot\left(\left((A_{2nom}\cdot\omega_{nom}-A_{1nom}\cdot\alpha_{nom})\cdot cos(\omega_{dnom}\cdot t)\right)+\left((-A_{2nom}\cdot\alpha_{nom}-A_{1nom}\cdot\omega_{dnom})\cdot sin(\omega_{dnom}\cdot t)\right)\right)$$

$$V_{nom}\ (\mathrm{t}) = e^{-\alpha_{nom}\cdot t}\cdot\left(A_{1nom}\cdot cos(\omega_{dnom}\cdot t)+A_{2nom}\cdot sin(\omega_{dnom}\cdot t)\right)$$

Upper Limit values for voltage (maximum), resistance (minimum), inductance (minimum), and capacitance (maximum):

$R_{min} = 0.95\ \Omega$ $\quad C_{max} = 1050\ p\mathrm{F}$ $\quad L_{min} = 4\ \mu\mathrm{H}$ $\quad V_{max} = 306000\mathrm{V}$

Calculations for alpha and omega based on Upper Limit values:

$$\alpha_{\mathrm{UL}} = \frac{R_{min}}{2\cdot L_{min}} \qquad \omega_{o\mathrm{UL}} = \frac{1}{\sqrt{L_{min}\cdot C_{max}}}$$

$$\alpha_{\mathrm{UL}} = (1.1875\cdot 10^5)\,\frac{1}{s} \qquad \omega_{o\mathrm{UL}} = (1.543\cdot 10^7)\frac{1}{s}$$

Calculations for damping frequency based on Upper Limit values of alpha and omega:

$$\omega_{dUL} = \sqrt{\frac{1}{L_{min}\cdot C_{max}}-\left(\frac{R_{min}}{2\cdot L_{min}}\right)^2}$$

$$\omega_{dUL} = (1.543\cdot 10^7)\frac{1}{s}$$

Calculations for A1 and A2 based on Upper Limit values:

$A_{1UL} = V_{max}$ $\qquad A_{1UL} = (3.06\cdot 10^5)V$

$A_{2UL} = \frac{\alpha_{UL}\cdot A_{1UL}}{\omega_{dUL}}$ $\qquad A_{2UL} = (2.355\cdot 10^3)V$

Double exponential equation for current & voltage waveforms based on Upper Limit value calculations:

$$i_{UL}(t) = \left((-c_{max})\cdot(e^{-\alpha_{UL}\cdot t})\right)\cdot\left(\left((A_{2UL}\cdot\omega_{dUL}-A_{1UL}\cdot\alpha_{UL})\cdot cos(\omega_{dUL}\cdot t)\right)+\left((-A_{2UL}\cdot\alpha_{UL}-A_{1UL}\cdot\omega_{dUL})\cdot sin(\omega_{dUL}\cdot t)\right)\right)$$

$$V_{UL}(t) = e^{-\alpha_{UL}\cdot t}\cdot\left(A_{1UL}\cdot cos(\omega_{dUL}\cdot t)+A_{2UL}\cdot sin(\omega_{dUL}\cdot t)\right)$$

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Lower Limit values for voltage (minimum), resistance (maximum), inductance (maximum), and capacitance (minimum):

$R_{max} = 4\ \Omega$ $C_{min} = 950\ pF$ $L_{max} = 20\ \mu H$ $V_{min} = 294000V$

Calculations for alpha and omega based on Lower Limit values:

$$\alpha_{LL} = \frac{R_{max}}{2 \cdot L_{max}} \qquad \omega_{oLL} = \frac{1}{\sqrt{L_{max} \cdot C_{min}}}$$

$$\alpha_{LL} = (1.0 \cdot 10^5)\frac{1}{s} \qquad \omega_{oLL} = (7.255 \cdot 10^6)\frac{1}{s}$$

Calculations for damping frequency based on Lower Limit values of alpha and omega:

$$\omega_{dLL} = \sqrt{\frac{1}{L_{max} \cdot C_{min}} - \left(\frac{R_{max}}{2 \cdot L_{max}}\right)^2}$$

$$\omega_{dLL} = (7.254 \cdot 10^6)\frac{1}{s}$$

Calculations for A1 and A2 based on nominal values:

$A_{1LL} = V_{min}$ $A_{1LL} = (2.94 \cdot 10^5)V$

$A_{2LL} = \frac{\alpha_{LL} \cdot A_{1LL}}{\omega_{dLL}}$ $A_{2LL} = (4.053 \cdot 10^3)V$

Double exponential equation for current & voltage waveforms based on Lower Limit values calculations:

$$i_{LL}(t) = \left((-c_{min}) \cdot (e^{-\alpha_{LL} \cdot t})\right) \cdot \left(\left((A_{2LL} \cdot \omega_{dLL} - A_{1LL} \cdot \alpha_{LL}) \cdot cos(\omega_{dLL} \cdot t)\right) + \left((-A_{2LL} \cdot \alpha_{LL} - A_{1LL} \cdot \omega_{dLL}) \cdot sin(\omega_{dLL} \cdot t)\right)\right)$$

$$V_{LL}(t) = e^{-\alpha_{LL} \cdot t} \cdot \left(A_{1LL} \cdot cos(\omega_{dLL} \cdot t) + A_{2LL} \cdot sin(\omega_{dLL} \cdot t)\right)$$

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CONCLUDING MATERIAL

Custodians:
Army – SY
Navy – SH
Air Force – 11
DISA – DC5

Preparing Activity:
DISA (DC5)
(Project EMCS-2022-021)

Review Activities:
Army - AC, AM, AR, AT, AV, CE, CR, GL, MD, MI, PT, TE
Navy – AS, CG, EC, MC, OS, YD
Air Force - 13, 19, 22
DTRA - DS

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